Qianfan-OCR: A Unified End-to-End Model for Document Intelligence

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https://github.com/baidubce/Qianfan-VL

Abstract

We present Qianfan-OCR, a 4B-parameter end-to-end document intelligence model that unifies document parsing, layout analysis, and document understanding within a single vision-language architecture. Unlike traditional multi-stage OCR pipelines that chain separate layout detection, text recognition, and language comprehension modules, Qianfan-OCR performs direct image-to-Markdown conversion and supports a broad range of prompt-driven tasks – from structured document parsing and table extraction to chart understanding, document question answering, and key information extraction – all within one model. A practical limitation of end-to-end OCR is the loss of explicit layout analysis, a capability that pipeline users routinely rely on for element localization and type classification. We introduce Layout-as-Thought to bridge this gap: an optional thinking phase triggered by ⟨think⟩ tokens, where the model generates structured layout representations (bounding boxes, element types, and reading order) before producing final outputs. This mechanism serves two purposes: (1) it recovers layout analysis functionality within the end-to-end paradigm, enabling users to obtain spatial grounding results directly; and (2) it provides targeted accuracy improvements on documents with complex layouts, cluttered elements, or non-standard reading orders, where structural priors help resolve recognition ambiguities. On OCR-specific benchmarks, Qianfan-OCR ranks first among all end-to-end models on OmniDocBench v1.5 (93.12) and OlmOCR Bench (79.8). It also achieves strong results on general OCR benchmarks including OCRBench (880), OCRBenchv2, and CCOCR, as well as document understanding tasks such as DocVQA, ChartQA, and CharXiv, matching general vision-language models of comparable scale. On public Key Information Extraction benchmarks, Qianfan-OCR achieves the highest average score, surpassing Gemini-3.1-Pro, Gemini-3-Pro, Seed-2.0, and Qwen3-VL-235B-A22B. The model is publicly accessible through Baidu AI Cloud Qianfan platform, with usage examples and best practices available at https://github.com/baidubce/Qianfan-VL.

1 Introduction

Current OCR systems face a three-way trade-off between cost, accuracy, and capability. *Traditional OCR pipelines* based on small specialized models offer low inference cost and high throughput, but require complex multi-stage preprocessing and postprocessing to handle diverse document layouts. *Specialized OCR large models* (Wei et al., 2024; 2025; Cui et al., 2025b; Poznanski et al., 2025) improve accuracy through two-stage architectures – layout detection followed by element-wise recognition – but introduce deployment complexity, inter-stage error propagation, and irreversible loss of visual context during text extraction. *General vision-language models* (Liu et al., 2024; Chen et al., 2024b) offer broad multimodal capabilities but incur higher inference costs and underperform specialized systems on structured document parsing tasks requiring precise layout preservation.

In practice, industrial OCR applications – document retrieval with chunking and indexing, contract review, key information extraction from receipts and certificates – often chain detection models, OCR models, and separate LLMs for downstream understanding. This fragmented approach increases deployment cost, limits end-to-end optimization, and requires careful orchestration of heterogeneous components.

We introduce Qianfan-OCR, a 4B-parameter unified end-to-end model that addresses these limitations with three key designs:

End-to-End Architecture: Qianfan-OCR integrates layout analysis, text recognition, and semantic understanding into a single vision-language model, eliminating inter-stage error propagation and enabling joint optimization across all tasks. The end-to-end design allows the model to retain full visual context throughout processing – spatial relationships, chart structures, and formatting that pipeline systems discard during text extraction. For tasks that do not require explicit layout analysis (e.g., simple document transcription or scene text recognition), the model directly outputs results without mandatory layout preprocessing.

Layout-as-Thought: A practical limitation of end-to-end OCR is the loss of explicit layout analysis –

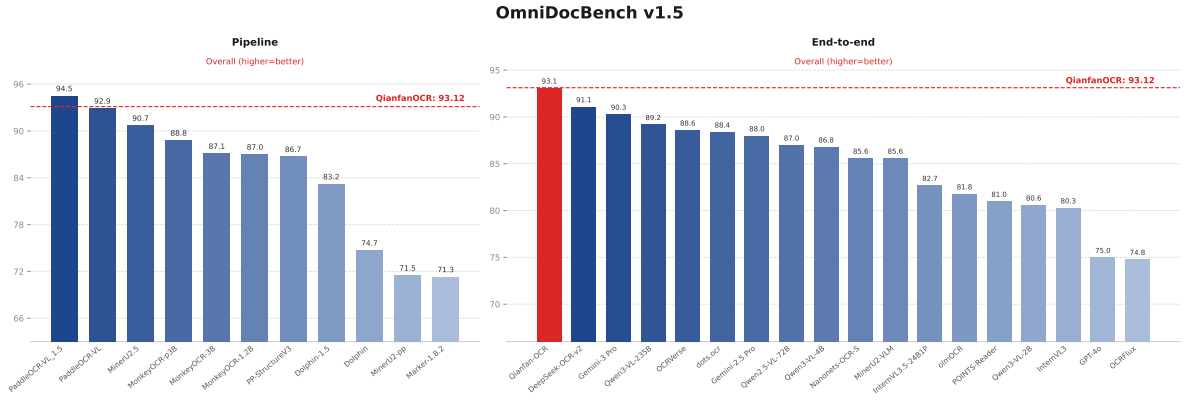

Figure 1: Performance on OmniDocBench v1.5 across pipeline (left) and end-to-end (right) models. Qianfan-OCR (red) achieves 93.12, ranking first among all end-to-end models. The red dashed line indicates Qianfan-OCR's score for cross-category comparison.

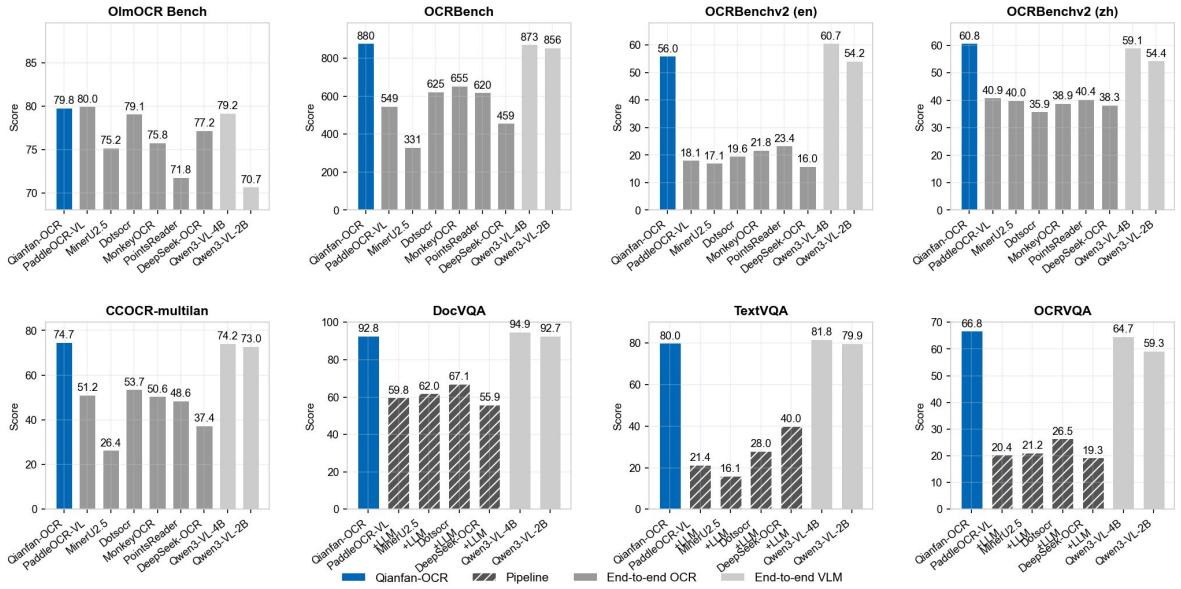

Figure 2: Performance on general OCR and document understanding benchmarks. **Top row:** OlmOCR Bench, OCRBench, OCRBenchv2 (en), and OCRBenchv2 (zh). **Bottom row:** CCOCR-multilan, and document understanding tasks (DocVQA, TextVQA, OCRVQA) where two-stage OCR+LLM pipelines (hatched bars) show significant degradation compared to end-to-end models.

a capability that pipeline systems inherently provide through dedicated detection modules. Layout-as-Thought recovers this within the end-to-end paradigm: an optional thinking phase triggered by ⟨think⟩ tokens, where the model generates bounding boxes, element types, and reading order before producing final outputs. This serves two purposes: (1) *functional* – users obtain structured layout results (element localization, type classification, spatial grounding) directly from an end-to-end model, bridging a key functionality gap relative to pipeline systems; (2) *enhancement* – the explicit structural priors help resolve ambiguities in documents with cluttered elements, complex multi-column layouts, or non-standard reading orders. For well-structured documents where the model already performs well, the layout phase is unnecessary; it targets the subset of challenging cases where structural reasoning provides measurable gains.

Unified OCR and Understanding: Beyond conventional OCR tasks (document parsing, handwriting recognition, table extraction), Qianfan-OCR handles cognitively demanding tasks including chart understanding, document question answering, and key information extraction – tasks requiring both precise text perception and semantic reasoning. Traditional OCR models lack comprehension capabilities, limiting them to character-level extraction; general VLMs possess reasoning abilities but underperform on structured parsing. Qianfan-OCR bridges this divide, combining OCR-specialist-level accuracy with document understanding capabilities in a single model controllable through prompts.

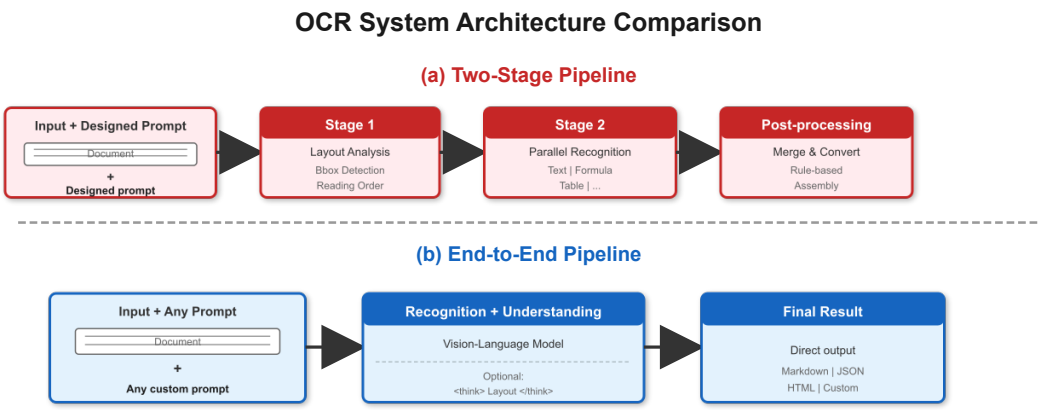

Figure 3: Architectural comparison between traditional two-stage OCR pipeline and Qianfan-OCR's end-to-end approach. **(a)** Traditional pipeline systems separate layout analysis and content recognition into independent stages, suffering from error propagation and irreversible loss of visual context. **(b)** Qianfan-OCR unifies all processing into a single vision-language model, accepting custom prompts for flexible task control and optionally generating intermediate layout reasoning via Layout-as-Thought ($\langle$think$\rangle$ tokens).

2 Related Work

We review three technical routes in OCR and position Qianfan-OCR relative to each.

Pipeline OCR Systems. Pipeline systems Cui et al. (2025a) decompose document parsing into layout detection, element-wise recognition, and rule-based assembly. Recent systems such as PaddleOCR-VL Cui et al. (2025b), MonkeyOCR, and MinerU 2.5 pair lightweight detection models with VLM-based recognizers, achieving strong accuracy with modular efficiency. Their key advantage is explicit layout analysis output (bounding boxes, element types), but they suffer from inter-stage error propagation and irreversible loss of visual context during text extraction. Qianfan-OCR recovers the layout analysis capability through Layout-as-Thought while avoiding the pipeline's cascading error problem.

End-to-End OCR Models. End-to-end approaches directly map document images to structured outputs. Nougat Blecher et al. (2023) demonstrated feasibility on academic papers; GOT-OCR 2.0 Wei et al. (2024) broadened format support (Markdown, LaTeX, TikZ) at 580M parameters; DeepSeek-OCR Wei et al. (2025) introduced context optical compression for efficiency; olmOCR Poznanski et al. (2025) scaled SFT-based training on large-scale web documents, while its successor olmOCR 2 further introduced GRPO reinforcement learning with unit-test rewards. More recently, Dolphin v2 proposed analyze-then-parse, and Logics-Parsing and Infinity-Parser explored layout-aware RL for structure prediction. These models primarily focus on recognition accuracy or efficiency but lack explicit layout analysis output – a functionality gap that Qianfan-OCR's Layout-as-Thought addresses. Qianfan-OCR relies on supervised fine-tuning with high-quality layout annotations, a complementary paradigm that future work could augment with reinforcement learning.

General Vision-Language Models. Large VLMs such as Qwen-VL Bai et al. (2023; 2025), InternVL Chen et al. (2024b); Zhu et al. (2025), and Gemini exhibit OCR capabilities as a byproduct of broad multimodal training, but are not optimized for structured document parsing: they incur higher inference costs, lack fine-grained layout control, and underperform specialized systems on structure-sensitive metrics (e.g., table TEDS, reading order accuracy). Qianfan-OCR targets OCR-specialist-level accuracy at comparable inference cost to these models, while additionally supporting explicit layout analysis and prompt-driven task control.

3 Model Architecture and Training Data

3.1 Architecture Overview

Qianfan-OCR adopts the multimodal bridging architecture from Qianfan-VL (Dong et al., 2025), consisting of three core components: a vision encoder for flexible visual encoding, a lightweight projection adapter for cross-modal alignment, and a language model backbone for text generation and reasoning. The overall architecture is illustrated in Figure 3(b).

Vision Encoder. The vision encoder employs **Qianfan-ViT**, pretrained as part of the Qianfan-VL frame-

work (Dong et al., 2025). It adopts the AnyResolution design that dynamically tiles input images into 448×448 patches, supporting variable-resolution inputs up to 4K. This is critical for OCR tasks where documents contain dense text, small fonts, and complex layouts that require high-resolution processing. The encoder consists of 24 Transformer layers with 1024 hidden dimensions, 16 attention heads, and a 14×14 patch size, producing 256 visual tokens per tile. With a maximum of 16 tiles, the encoder can represent a single document image with up to 4,096 visual tokens, providing sufficient spatial resolution for fine-grained character recognition.

Language Model Backbone. We adopt **Qwen3-4B** (Bai et al., 2025) as the language model backbone. The model has 4.0B total parameters (3.6B non-embedding), 36 layers, 2560 hidden dimensions, and a 32K native context window (extendable to 131K via YaRN). This scale strikes a balance between reasoning capability and deployment efficiency – large enough for complex document understanding and layout reasoning, yet practical for single-GPU serving in production. The model uses Grouped-Query Attention (GQA) (Ainslie et al., 2023) with 32 query heads and 8 KV heads, reducing KV cache memory by 4× compared to standard multi-head attention while maintaining generation quality. RMSNorm (Zhang and Sennrich, 2019) is used for layer normalization, improving training stability.

Cross-Modal Adapter. A lightweight two-layer MLP with GELU activation bridges the vision encoder and the language model, projecting visual features from the encoder's representation space (1024 dimensions) into the language model's embedding space (2560 dimensions). This simple design minimizes adapter parameters while ensuring effective cross-modal alignment. During Stage 1 training, only the adapter is trained with a higher learning rate for fast alignment, while subsequent stages perform full-parameter training.

3.2 Large-Scale Data Synthesis Pipelines

We develop six data synthesis pipelines covering document parsing, key information extraction, complex tables, chart understanding, formula recognition, and multilingual OCR.

Document Parsing Data Synthesis: We construct an automated pipeline that converts document images into structured Markdown using PaddleOCR-VL (Cui et al., 2025b) for layout detection and content recognition, with bbox coordinates normalized to [0, 999] for resolution invariance. Tables are converted to HTML via OTSL intermediate format, and formulas are wrapped in $$ blocks. Automatic filtering removes repetitive or extreme-length samples, and image-level augmentations (compression, flipping, blur) improve robustness.

A key design choice is the layout label system. We compare PaddleOCR-VL and MinerU 2.5 along label granularity and detection accuracy. The main difference lies in body text labels: PaddleOCR-VL provides fine-grained categories (`text`, `vertical_text`, `paragraph_title`, `doc_title`, `abstract`, `content`, `reference`, `reference_content`, `aside_text`), while MinerU 2.5 uses coarser labels (`text`, `title`, `list`, `aside_text`). Finer granularity directly benefits downstream tasks – e.g., distinguishing `abstract` from `content` supports structured extraction from papers, and separating `reference` enables clean bibliography parsing. We also evaluate both systems on a multi-type document layout benchmark, where PaddleOCR-VL achieves consistently higher detection accuracy. We therefore adopt PaddleOCR-VL's label system and use it as the annotation engine. Our final taxonomy contains 25 categories in four groups: *text elements* (12 labels), *headers/footers* (4), *figures/tables* (6), and *formulas* (3).

Layout-as-Thought Data Construction: We construct training data where the model generates structured layout analysis within `<think>` tokens before final output, listing bbox coordinates, element labels, and content summaries as intermediate reasoning enclosed in `<layout>...</layout>` tags. Users activate this by appending `<think>` tokens to queries. The layout phase focuses model attention on relevant document regions before generation, improving performance on documents requiring spatial reasoning (complex layouts, multi-column text, interleaved figures).

Key Information Extraction (KIE): For KIE tasks, we construct datasets for two scenarios: complete extraction ("what you see is what you get") and targeted extraction (user-specified keys). To address hallucination in teacher models, we combine open-source data with small model pre-annotations for multi-model collaborative labeling. We implement semantic generalization for keys across different regions and formats, constructing multiple synonymous descriptions for the same field. The pipeline includes quality enhancement through direction correction and image enhancement for low-resolution inputs, hard rule filtering using business logic (e.g., verifying "unit price × quantity = total"), and difficult sample mining for long sequences with 5+ detail rows and dense text documents. Sample distribution is rebalanced based on task difficulty to enhance stability in extreme scenarios.

Complex Tables: We combine programmatic synthesis with real document extraction. The programmatic pipeline randomly generates tables with 3-20 rows/columns supporting random cell merging, populates

content via Faker library or LLMs covering diverse data types, randomly samples from 50+ professional CSS themes, renders via Jinja2 and KaTeX engines, and applies geometric transformations, color perturbations, and blur augmentations. For real document tables, we use internal parsing tools to detect and extract table regions, parse with both PaddleOCR-VL and internal table models, convert both outputs to HTML, and perform consistency validation to filter samples with significant structural or content differences, ensuring reliable annotations while preserving real document layout and noise characteristics.

Chart Understanding: We build an automated synthesis pipeline based on arXiv LaTeX sources (2022-present). The pipeline systematically extracts Figure code blocks through rule parsing, re-renders using TexLive engine to obtain lossless vector images, leverages caption parameters as ground truth, and uses VLMs to generate detailed visual descriptions capturing visual encoding, statistical features, spatial layout, and fine-grained distribution characteristics. We categorize 11 mainstream chart types and construct "metadata + visual description" driven synthesis. Custom reasoning tasks are designed for different chart types: trend analysis for line charts, correlation and distribution features for scatter plots, outlier detection for box plots. This pipeline synthesizes over 300,000 high-accuracy, diverse multimodal instruction-tuning samples.

Multilingual OCR Data Construction: To extend language coverage to 192 languages, we adopt a *reverse synthesis* approach starting from the HPLT multilingual corpus. The pipeline performs text-font renderability filtering using fonttools character set validation, then renders document images with differentiated handling for different writing systems (Latin, Cyrillic, Arabic, South Asian, Southeast Asian, Han). Key features include automatic RTL text direction detection, Arabic character reshaping, and word-level line breaking. Diverse typesetting variations (font size, column layout, margins, spacing, texture backgrounds) are randomized to approximate real document distributions.

Document Image Augmentation: We employ two augmentation pipelines: one for OCR tasks (allowing mild geometric perturbations) and one for layout parsing tasks (preserving geometric consistency). Both apply three noise stages: (1) *text noise* (broken strokes, ink bleeding, character misalignment), (2) *background noise* (texture, color drift, watermarks), and (3) *imaging noise* (blur, moiré, shadows, exposure variation). Additionally, rotation augmentation (90°, 180°, 270°, and ±15°) significantly improves performance on KIE and table recognition tasks where documents frequently appear in non-standard orientations.

Through these specialized synthesis pipelines, we generate large-scale, high-quality training data covering diverse OCR scenarios, providing comprehensive data support for Qianfan-OCR's multi-stage progressive training.

4 Training Recipe

Qianfan-OCR adopts the proven multi-stage progressive training methodology from Qianfan-VL (Dong et al., 2025), which systematically builds model capabilities from basic cross-modal alignment through advanced reasoning tasks. The key adaptation for OCR scenarios lies in the **data mixture composition**, where we significantly enhance OCR-specific domains while maintaining the overall training framework. The training pipeline consists of four stages with carefully designed data distributions optimized for document intelligence.

Stage 1: Cross-Modal Alignment (50B tokens) – Establishes fundamental vision-language alignment with adapter-only training, using basic image-caption pairs and simple OCR tasks to ensure stable initialization.

Stage 2: Foundational OCR Training (2T tokens) – Develops comprehensive OCR capabilities through full parameter training with OCR-heavy data mixture: Document OCR (45%), Scene OCR (25%), Caption (15%), and Specialized OCR tasks including handwriting, formulas, tables, and multilingual text (15%).

Stage 3: Domain-Specific Enhancement (800B tokens) – Implements targeted enhancement for enterprise-critical OCR domains with balanced mixture: Complex Tables (22%), Formula Recognition (20%), Chart Understanding (18%), Information Extraction (18%), Multilingual OCR (12%), and Document Understanding (10%). Maintains 70% domain-specific data and 30% general data to enhance specialization while preventing catastrophic forgetting.

Stage 4: Instruction Tuning and Reasoning Enhancement (millions of instruction samples) – Covers a comprehensive set of document intelligence tasks including document parsing, layout analysis, handwriting recognition, scene text recognition, formula recognition, table recognition, multi-page document parsing, chart QA, document QA, and complex table QA. The instruction data is constructed through three complementary strategies: (1) *Public data curation*: we collect publicly available OCR-related training datasets and perform instruction rewriting and generalization using DeepSeek models to diversify prompt styles and task formulations; (2) *Reverse synthesis*: for tasks amenable to reverse generation (e.g., ta-

bles, exam papers), we construct large-scale QA pairs by generating questions conditioned on structured ground-truth content; (3) *Chart data mining*: we extract chart-figure pairs from a large corpus of academic papers via their LaTeX source code and generate chart understanding QA pairs grounded in the original source, significantly enhancing chart comprehension capabilities. All instruction data undergoes systematic prompt generalization and rewriting to improve robustness to diverse user instructions.

The critical differentiator from general VLM training lies in the **OCR-centric data composition** throughout all stages, with particular emphasis on document parsing, table/chart understanding, and information extraction tasks. Detailed data synthesis pipelines for each domain are described in Section 3.

Training Infrastructure and Iteration Strategy. All training is conducted on **1,024 Baidu Kunlun P800 chips** using 3D parallelism (data, tensor, and pipeline parallelism with communication-computation overlap), processing over 2.85T tokens across all stages. In practice, Stages 1 and 2 are trained once to establish a stable foundation checkpoint, while Stages 3 and 4 are iterated multiple times to explore different domain-specific data mixtures, sampling ratios, and instruction tuning configurations. Since Stages 3 and 4 account for a smaller token budget (800B + instruction tuning vs. 2T for Stage 2), each iteration completes quickly. The full four-stage pipeline completes **within a week**, and Stage 3/4 iterations take approximately one day each, supporting systematic ablation and optimization of OCR-specific training recipes.

4.1 Hyperparameters

Table 1 summarizes the key hyperparameters for each training stage.

Hyperparameter	**Stage 1**	**Stage 2**	**Stage 3**	**Stage 4**
Trainable modules	Adapter	All	All	All
Tokens	50B	2T	800B	–
Global batch size (samples)	1024	2048	2048	512
Peak learning rate	1e-3	2e-5	1e-5	1e-5
LR schedule	Cosine	Cosine	Cosine	Cosine
Warmup ratio	0.01	0.01	0.01	0.03
Optimizer	AdamW (β_1=0.9, β_2=0.95)			
Weight decay	0.05			
Max sequence length	32768			

Table 1: Training hyperparameters for each stage. Stage 1 trains only the adapter with a higher learning rate for fast alignment. Stages 2–4 use full-parameter training with lower learning rates.

4.2 Ablation Study: Multi-Stage Training Effectiveness

Prior to training the 4B Qianfan-OCR model, we conduct low-cost ablation studies on Qianfan-VL-8B – a model from the same architectural family that has undergone large-scale general-purpose continual pretraining – to systematically validate our multi-stage training recipe. Starting from a Stage 1 (adapter-only alignment) checkpoint, we evaluate the contribution of each subsequent training stage to OCR performance. All configurations employ full-parameter training, which prior Qianfan-VL experiments have confirmed to be essential for effective domain transfer. Results are summarized in Table 2.

Stage 1	**Stage 2**	**Stage 3**	**Stage 4**	**AVG**
✓	–	–	✓	71.37
✓	–	OCR	✓	75.97
✓	–	OCR + General	✓	80.07
✓	General	–	✓	83.47
✓	General	OCR	✓	84.09
✓	General	OCR + General	✓	**84.39**

Table 2: Ablation study on multi-stage training effectiveness using Qianfan-VL-8B. "OCR" denotes domain-specific OCR data, "General" denotes general-purpose data, and "OCR + General" denotes a 1:1 mixture. Stage 4 is instruction tuning with a fixed set of alignment samples across all configurations. Average accuracy is computed over multiple OCR benchmarks.

Stage 2 (foundational pretraining) is essential. Skipping Stage 2 and proceeding directly from Stage 1 to domain-specific Stage 3 training yields significantly lower performance. Even the best Stage 3-only configuration (OCR + General mixture, 80.07%) falls substantially short of Stage 2 followed by Stage 4 alone

(83.47%), indicating that large-scale general-purpose pretraining provides a critical capability foundation that cannot be substituted by domain-specific data alone.

Domain-specific enhancement benefits from general data mixing. In Stage 3, a 1:1 mixture of OCR-specific and general-purpose data (80.07%) consistently outperforms pure OCR data (75.97%), suggesting that maintaining general capability during domain specialization acts as an effective regularizer and prevents overfitting to narrow OCR patterns. The same trend holds when Stage 2 is included: OCR + General in Stage 3 (84.39%) outperforms OCR-only Stage 3 (84.09%).

The complete pipeline achieves optimal performance. The full four-stage configuration (Stage 1 → Stage 2 General → Stage 3 OCR+General → Stage 4) achieves the highest accuracy of 84.39%, representing a +13.02% absolute improvement over the Stage 1+4 baseline. Notably, this configuration also surpasses Qwen2.5-VL-7B (79.30%) by +5.09%, despite the latter being a strong general-purpose VLM. In practice, we keep the Stage 4 data volume fixed and incorporate all additional domain-specific data augmentation into Stage 3, which proves to be the most effective strategy for scaling OCR capabilities without compromising the general abilities established in Stage 2.

These findings, obtained at relatively low cost on the 8B model, directly motivated the training recipe adopted for the 4B Qianfan-OCR. The consistency of improvements across training stages suggests that this progressive recipe generalizes across model scales within the same architectural family.

5 Evaluation Framework

To comprehensively assess Qianfan-OCR's capabilities across the spectrum from specialized OCR to document understanding, we employ a multi-dimensional evaluation framework spanning four key categories:

Specialized OCR Model Benchmarks: We evaluate OCR-specific capabilities using benchmarks designed for specialized document parsing systems: Omni-Doc-Bench v1.5 (Ouyang et al., 2024) for diverse PDF document parsing, OLMOCRBench (AllenAI, 2024) for end-to-end document OCR evaluation, CCOCR for multilingual OCR recognition across diverse scripts, and BigDocs (Rodriguez et al., 2024) for large-scale document processing tasks. These benchmarks assess the model's ability to achieve recognition accuracy competitive with specialized OCR systems.

General OCR Capability Benchmarks: To evaluate general-purpose optical character recognition across diverse scenarios, we use OCRBench (Chen et al., 2024a) and OCRBench v2, comprehensive benchmarks covering scene text, document text, handwriting recognition, formula recognition, and multilingual text across various real-world conditions. These evaluations ensure the model maintains robust OCR performance beyond specialized document parsing scenarios.

Document Understanding Benchmarks: We assess document comprehension and reasoning capabilities using a diverse set of understanding-oriented benchmarks: TextVQA (Singh et al., 2019) for scene text question answering, DocVQA (Mathew et al., 2021) for document visual question answering, CharXiv document question (CharXiv_DQ) and reasoning question (CharXiv_RQ) tasks (Wang et al., 2024) for academic document understanding, ChartQA (Masry et al., 2022) and ChartQAPro (Masry et al., 2025) for chart interpretation, and ChartBench for comprehensive chart reasoning evaluation. These benchmarks evaluate the model's ability to perform high-level semantic understanding and reasoning over visual document content, including complex table analysis and chart interpretation.

Key Information Extraction (KIE): We evaluate KIE capabilities across five public benchmarks: OCRBench KIE (Chen et al., 2024a) (structured field extraction from receipts, invoices, and forms), OCRBenchv2 KIE in both English and Chinese (cross-lingual extraction across diverse document templates), CCOCR KIE (Chinese document field extraction including ID cards, business licenses, and financial documents), and Nanonets KIE (real-world invoice and receipt parsing, measured by F1 score). All scores are normalized to a 0–100 scale. Since specialized OCR models lack native KIE capabilities, we compare against commercial large models (Gemini-3.1-Pro, Gemini-3-Pro, Seed-2.0) and open-source VLMs (Qwen3-4B-VL, Qwen3-VL-235B-A22B).

Comparative Evaluation Against Pipeline Systems: To fairly compare against traditional OCR-then-LLM approaches, we establish baseline systems that combine specialized OCR models with language models of comparable parameter count to Qianfan-OCR. This comparative evaluation on understanding benchmarks isolates the architectural effects from parameter count differences, providing controlled evidence for the trade-offs between end-to-end and pipeline approaches.

6 Experimental Results

We conduct comprehensive evaluations across OCR-specific benchmarks, general OCR benchmarks, and document understanding tasks. Our evaluation framework is primarily based on VLMEvalKit (Duan et al., 2024) with modifications tailored to different benchmark requirements.

Evaluation Methodology. For OCR-specific benchmarks, we follow official metrics or adopt results from recent publications. Specifically, we reference metrics from the OmniDocBench official leaderboard (Ouyang et al., 2024), DeepSeek-OCR v2 (Wei et al., 2026), and PaddleOCR-VL 1.5 (Cui et al., 2026) papers. For benchmarks with official evaluation scripts (e.g., OmniDocBench v1.5), we directly reuse their implementations.

Model Categorization. We categorize comparison models based on their architectural paradigms:

- **Pipeline OCR Systems:** Models that perform layout analysis first, followed by parallel text recognition (e.g., PaddleOCR-VL, MonkeyOCR series, Dolphin series).
- **End-to-end Models:** Vision-language models that directly process images through prompts without separate detection stages, including both specialized OCR models (e.g., MinerU2.5, Dotsocr, DeepSeek-OCR) and general VLMs (e.g., Qwen3-VL, GPT-4o, Gemini).

Benchmark-Specific Considerations. For general OCR benchmarks (OCRBench, OCRBenchv2, CCOCR-multilan), we compute metrics using VLMEvalKit. Since specialized OCR models typically do not report these metrics, we integrate their OCR outputs in our evaluation environment. Note that these benchmarks include not only pure OCR recognition but also understanding and key information extraction (KIE) tasks, where specialized OCR models generally underperform. For document understanding tasks, specialized OCR models cannot directly complete such tasks (metrics approach zero). To simulate real-world usage scenarios, we employ a two-stage pipeline: first using specialized OCR models for text extraction, then feeding the extracted text to Qwen3-4B LLM for answer generation.

6.1 OCR-Specific Benchmarks

Model	OlmOCR Bench	ArXiv	Old scans math	Tables	Old scans	Headers & footers	Multi column	Long tiny text	Base
Pipeline									
PaddleOCR-VL	**80.0**	**85.7**	**71.0**	**84.1**	**37.8**	**97.0**	79.9	85.7	98.5
PaddleOCR-VL-1.5	79.1	84.6	63.8	81.8	36.9	95.5	**80.9**	**90.5**	**98.9**
MonkeyOCR	75.8	83.8	68.8	74.6	36.1	91.2	76.6	80.1	95.3
End-to-end Model									
MinerU2.5	75.2	76.6	54.6	84.9	33.7	**96.6**	78.2	83.5	93.7
Dotsocr	79.1	**82.1**	64.2	**88.3**	40.9	94.1	82.4	81.2	99.5
PointsReader	71.8	81.0	74.0	86.0	36.5	19.6	**85.2**	**92.8**	99.1
DeepSeek-OCR	77.2	77.2	73.6	80.2	33.3	96.1	66.4	79.4	**99.8**
Qwen3-VL-4B	79.2	81.9	**76.6**	79.3	40.7	88.8	78.8	88.0	99.1
Qwen3-VL-2B	70.7	78.1	72.5	69.3	38.0	51.2	76.1	82.1	98.6
Qianfan-OCR (Ours)	**79.8**	80.1	73.1	81.6	**42.0**	92.2	80.4	89.1	99.6

Table 3: Detailed performance on OlmOCR Bench across document categories. Models are categorized into Pipeline (two-stage) and End-to-end architectures. Best results in each section are in **bold**.

Table 3 presents detailed performance on OlmOCR Bench across document categories. Qianfan-OCR achieves the highest overall score (79.8) among end-to-end models, competitive with the top pipeline system PaddleOCR-VL (80.0), with strong scores on Base (99.6), Headers & footers (92.2), and Old scans (42.0, best among end-to-end).

Table 4 presents results on OmniDocBench v1.5. Among end-to-end models, Qianfan-OCR achieves the highest overall score of 93.12, surpassing DeepSeek-OCR-v2 (91.09), Gemini-3 Pro (90.33), and all other end-to-end models across every sub-metric. Notably, Qianfan-OCR also outperforms several pipeline systems including MinerU2.5 (90.67) and MonkeyOCR-pro-3B (88.85), narrowing the gap with the top pipeline system PaddleOCR-VL 1.5 (94.50).

Analysis of Layout-as-Thought (Thinking Mode). We additionally evaluate Qianfan-OCR with the Layout-as-Thought mechanism enabled at inference time (denoted Qianfan-OCR-think). Compared to the default mode, the thinking variant achieves an overall score of 92.64 vs. 93.12, with per-metric results: $\text{Text}^{\text{Edit}}$ 0.052 vs. 0.041, $\text{Formula}^{\text{CDM}}$ 91.92 vs. 92.43, $\text{Table}^{\text{TEDs}}$ 91.21 vs. 91.02, $\text{Table}^{\text{TEDss}}$ 94.03 vs. 93.85, and $\text{R-order}^{\text{Edit}}$ 0.051 vs. 0.049. While the aggregate score is slightly lower, the thinking variant

Model	Overall ↑	Text^{Edit} ↓	Formula^{CDM} ↑	Table^{TEDs} ↑	Table^{TEDss} ↑	R-order^{Edit} ↓
Pipeline						
Marker-1.8.2	71.30	0.206	76.66	57.88	71.17	0.250
MinerU2-pp	71.51	0.209	76.55	70.90	79.11	0.225
Dolphin	74.67	0.125	67.85	68.70	77.77	0.124
Dolphin-1.5	83.21	0.092	80.78	78.06	84.10	0.080
PP-StructureV3	86.73	0.073	85.79	81.68	89.48	0.073
MonkeyOCR-pro-1.2B	86.96	0.084	85.02	84.24	89.02	0.130
MonkeyOCR-3B	87.13	0.075	87.45	81.39	85.92	0.129
MonkeyOCR-pro-3B	88.85	0.075	87.25	86.78	90.63	0.129
MinerU2.5	90.67	0.047	88.46	88.22	92.38	0.044
PaddleOCR-VL	92.86	0.035	91.22	90.89	94.76	0.043
PaddleOCR-VL 1.5	**94.50**	**0.035**	**94.21**	**92.76**	**95.79**	**0.042**
End-to-end Model						
OCRFlux	74.82	0.193	68.03	75.75	80.23	0.202
GPT-4o	75.02	0.217	79.70	67.07	76.09	0.148
InternVL3	80.33	0.131	83.42	70.64	77.74	0.113
Qwen3-VL-2B	80.58	0.103	82.32	69.71	74.14	0.113
POINTS-Reader	80.98	0.134	79.20	77.13	81.66	0.145
olmOCR	81.79	0.096	86.04	68.92	74.77	0.121
InternVL3.5-241B	82.67	0.142	87.23	75.00	81.28	0.125
MinerU2-VLM	85.56	0.078	80.95	83.54	87.66	0.086
Nanonets-OCR-s	85.59	0.093	85.90	80.14	85.57	0.108
Qwen3-VL-4B	86.78	0.055	86.55	79.29	84.26	0.084
Qwen2.5-VL-72B	87.02	0.094	88.27	82.15	86.22	0.102
Gemini-2.5 Pro	88.03	0.075	85.82	85.71	90.29	0.097
dots.ocr	88.41	0.048	83.22	86.78	90.62	0.053
OCRVerse	88.56	0.058	86.91	84.55	88.45	0.071
Qwen3-VL-235B	89.15	0.069	88.14	86.21	90.55	0.068
Gemini-3 Pro	90.33	0.065	89.18	88.28	90.29	0.071
DeepSeek-OCR-v2	91.09	0.048	90.31	87.75	92.06	0.057
Qianfan-OCR (Ours)	**93.12**	**0.041**	**92.43**	**91.02**	**93.85**	**0.049**

Table 4: Performance comparison on OmniDocBench. Models are categorized into Pipeline (two-stage) and End-to-end architectures. ↑ indicates higher is better, ↓ indicates lower is better. Best results in each section are in **bold**. Data source: https://github.com/opendatalab/OmniDocBench

shows improvements on table-related metrics (Table^{TEDs}: +0.19, Table^{TEDss}: +0.18). More importantly, per-sample analysis reveals that Layout-as-Thought provides targeted benefits on structurally complex documents but can be counterproductive on simpler ones.

To investigate this, we sort all OmniDocBench v1.5 samples by their layout label entropy in descending order and plot the cumulative score as samples are progressively included (Figure 4). Layout label entropy measures the diversity of element types (text, table, formula, figure, etc.) on each page – higher entropy indicates more heterogeneous layouts.

The results reveal a clear pattern: in the high-entropy region (left portion of the curve), where documents contain diverse element types such as mixed text, formulas, tables, and figures, enabling thinking provides a consistent score advantage. As lower-entropy samples are progressively included – documents with more homogeneous layouts (e.g., pure text pages) – the gap narrows and eventually reverses, with the no-think mode achieving a higher cumulative total. This indicates that Layout-as-Thought introduces unnecessary overhead on structurally simple documents, where explicit layout reasoning provides no additional benefit and may even interfere with direct recognition.

In practice, users should decide whether to enable the thinking mode based on the layout complexity of their target documents: for heterogeneous pages with mixed element types (exam papers, technical reports, newspapers), enabling thinking improves accuracy; for homogeneous documents (single-column text, simple forms), disabling thinking yields better results with lower latency.

Illustrative Example. Figure 5 presents a concrete example of Layout-as-Thought on a math exam paper with mixed content (text, formulas, geometric diagrams, and multi-column layout). When the `<think>` token is activated, the model first generates a structured layout analysis enumerating each document element in reading order:

```
<think>
<box>[[<COORD_779>,<COORD_057>,<COORD_890>,<COORD_071>]]</box>
```

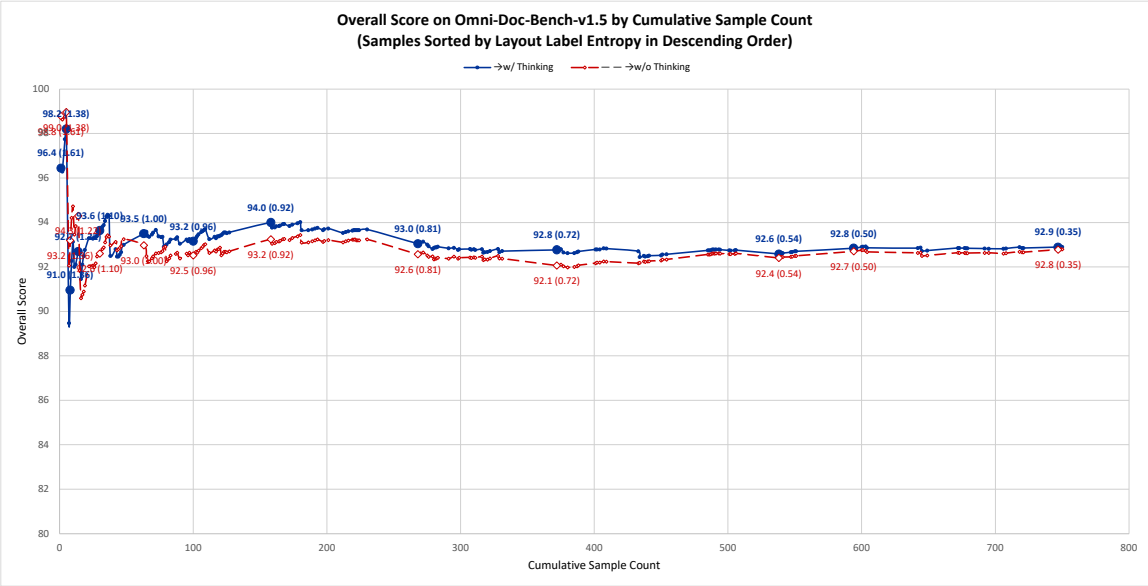

Figure 4: Cumulative OmniDocBench v1.5 score with samples sorted by layout label entropy (descending). In the high-entropy region (left), enabling thinking provides a stable advantage. As lower-entropy samples are included, the gap narrows and eventually reverses, with the no-think mode achieving a higher total score overall.

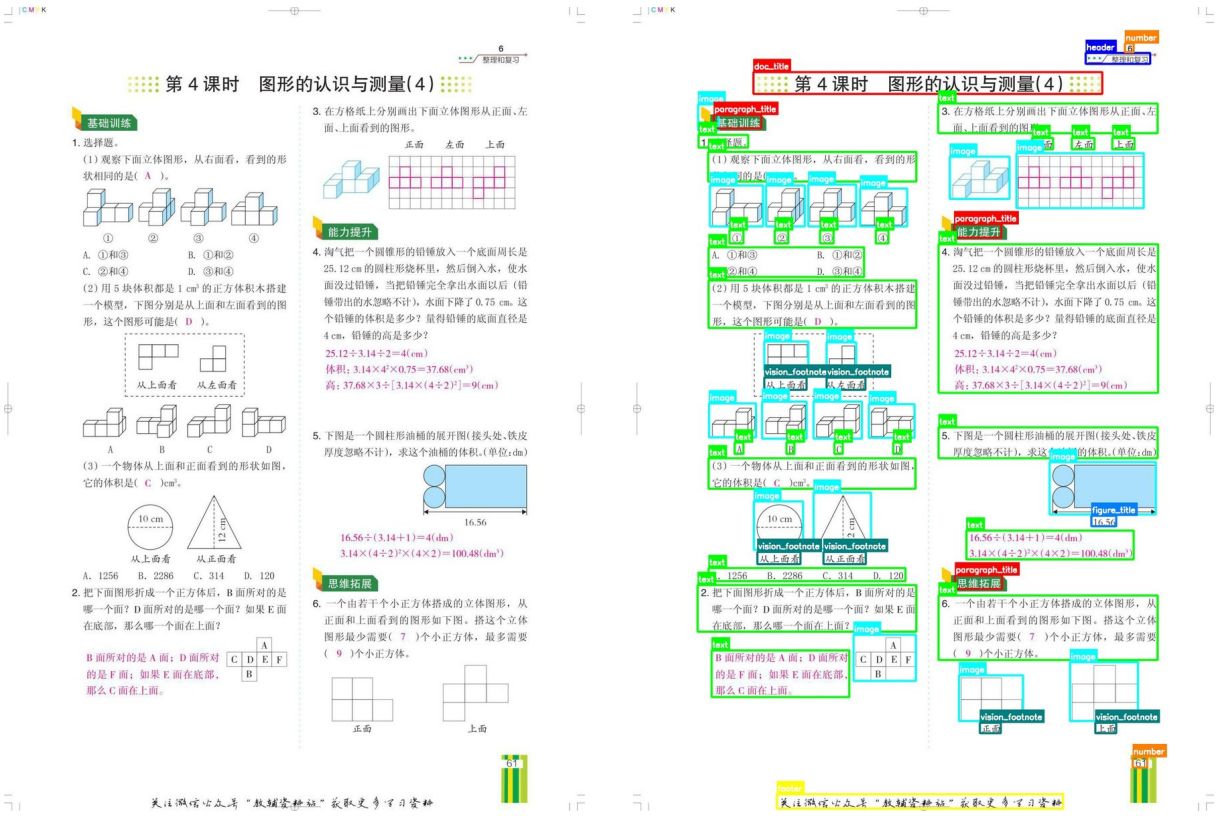

Figure 5: Layout-as-Thought example on a math exam paper. **Left:** Original document image. **Right:** Visualization of bounding boxes generated during the thinking phase, with element types color-coded (e.g., text, image, paragraph_title, vision_footnote).

```
<label>header</label>
<brief>Section title indicating review and organization</brief>
<box>[[<COORD_207>,<COORD_081>,<COORD_808>,<COORD_108>]]</box>
<label>doc_title</label>
<brief>Lesson title: Graphical Recognition and Measurement (4)</brief>
<box>[[<COORD_139>,<COORD_134>,<COORD_221>,<COORD_151>]]</box>
<label>paragraph_title</label>
<brief>Subsection title: Basic Training</brief>
<box>[[<COORD_129>,<COORD_180>,<COORD_488>,<COORD_217>]]</box>
<label>text</label>
<brief>Asks to identify shapes seen from the right side...</brief>
<box>[[<COORD_132>,<COORD_222>,<COORD_224>,<COORD_273>]]</box>
<label>image</label>
<brief>A 3D arrangement of white cubes with light blue edges...</brief>
  ...  (60+ layout elements in reading order)
</think>
```

Each entry contains three fields: a `<box>` with normalized bounding box coordinates, a `<label>` indicating the element type (from 25 layout categories including *text*, *image*, *paragraph_title*, *display_formula*, *table*, *vision_footnote*, etc.), and a `<brief>` providing a concise content summary for text-type elements. The model then generates the final OCR response conditioned on this structured layout reasoning.

Coordinate Special Tokens. All bounding box coordinates are normalized to [0, 999] and represented as dedicated special tokens `<COORD_0>` through `<COORD_999>`. These tokens are introduced during Stage 3 continued pretraining alongside layout analysis data, enabling the model to learn spatial representations efficiently. Compared to encoding coordinates as plain digit sequences (e.g., "779" requires 3 tokens), each coordinate consumes only a single token, reducing the thinking output length by approximately 50% and substantially decreasing inference latency for the layout reasoning phase. This compact representation is critical for practical deployment, as complex documents may contain 60+ layout elements in a single page.

How Layout-as-Thought Guides Response Generation. The structured thinking output benefits the final response in two key ways: (1) *Element-type-aware generation*: by explicitly identifying element categories (formula, table, image, text), the model applies appropriate rendering formats in the response – wrapping mathematical content in $$ ... $$ blocks, converting table structures to HTML, and inserting image placeholders at correct positions; (2) *Reading-order-guided sequencing*: the thinking phase enumerates elements following the natural reading order of the document (handling multi-column layouts, interleaved figures, and footnotes), providing an explicit ordering signal that the response generation can follow to produce correctly sequenced output.

6.2 General OCR Benchmarks

Table 5 shows performance on general OCR benchmarks covering diverse scenarios including scene text, documents, handwriting, and multilingual text. Qianfan-OCR achieves the highest OCRBench score (880) among all models and demonstrates strong performance on OCRBenchv2 (56.0 en / 60.77 zh), CCOCR-multilan (76.7), and CCOCR-overall (79.3). Compared with the similarly-sized general-purpose VLM Qwen3-VL-4B, Qianfan-OCR surpasses it on OCRBench (880 vs. 873) while showing a modest gap on OCRBenchv2 en (56.0 vs. 60.68), and leading on both CCOCR metrics (76.7 vs. 74.2 multilan, 79.3 vs. 76.5 overall). This trade-off is expected: Qianfan-OCR prioritizes specialized OCR capabilities (OmniDocBench 93.12, OlmOCR 79.8, both ranking first among all end-to-end models) while retaining competitive general performance without significant degradation.

Model	OCRBench	OCRBenchv2 (en/zh)	CCOCR-multilan	CCOCR-overall
Qianfan-OCR (Ours)	**880**	56.0 / **60.77**	**76.7**	**79.3**
PaddleOCR-VL	549	18.15 / 40.86	45.5	29.1
MinerU2.5	331	17.09 / 40.03	43.2	24.4
Dotsocr	625	19.59 / 35.94	47.2	34.6
MonkeyOCR	655	21.78 / 38.91	43.8	35.2
PointsReader	620	23.40 / 40.39	43.5	35.0
DeepSeek-OCR	459	15.98 / 38.31	32.5	27.6
Qwen3-VL-4B	873	**60.68** / 59.13	74.2	76.5
Qwen3-VL-2B	856	54.18 / 54.41	73.0	71.5

Table 5: Performance on general OCR benchmarks. Qianfan-OCR achieves competitive scores while maintaining strong understanding capabilities (Table 6).

Notable observations:

- **OCRBench:** Qianfan-OCR scores 880, surpassing Qwen3-VL-4B (873) and ranking first among all models
- **OCRBenchv2:** Qwen3-VL-4B leads on English (60.68 vs. 56.0), while Qianfan-OCR achieves the best Chinese text recognition (60.77), significantly outperforming all specialized OCR models on both languages
- **CCOCR:** Qianfan-OCR leads on both CCOCR-multilan (76.7 vs. 74.2) and CCOCR-overall (79.3 vs. 76.5) over Qwen3-VL-4B, demonstrating strong multilingual and comprehensive OCR capabilities

6.3 Document Understanding Benchmarks

Document understanding benchmarks evaluate the model's ability to perform visual question answering, chart interpretation, and reasoning tasks requiring both accurate text perception and semantic comprehension.

Table 6 compares end-to-end models with two-stage OCR+LLM systems, where specialized OCR models first extract text and Qwen3-4B then generates answers from the extracted text.

Benchmark	End-to-End Models			Two-Stage OCR+LLM Systems			
	Qianfan -OCR	**Qwen3 -VL-4B**	**Qwen3 -VL-2B**	**PaddleOCR +Qwen3-4B**	**MinerU2.5 +Qwen3-4B**	**Dotsocr +Qwen3-4B**	**DeepSeek OCR +Qwen3-4B**
OCRVQA_TESTCORE	**66.8**	64.7	59.3	20.4	21.2	26.5	19.3
TextVQA_VAL	80.0	**81.8**	79.9	21.4	16.1	28.0	40.0
DocVQA	92.8	**94.9**	92.7	59.8	62.0	67.1	55.9
CharXiv_DQ	**94.0**	81.8	69.7	0.0	0.0	0.0	0.0
CharXiv_RQ	**85.2**	48.5	41.3	0.0	0.0	0.0	0.0
ChartQA_TEST	**88.1**	83.3	78.3	56.8	7.0	14.6	23.4
ChartQAPro	**42.9**	36.2	24.5	20.9	18.7	18.4	18.1
ChartBench	**85.9**	74.9	73.2	17.2	5.7	2.0	6.1

Table 6: End-to-end models vs. two-stage OCR+LLM systems on document understanding benchmarks. Two-stage systems use Qwen3-4B as the downstream LLM. Best end-to-end results in **bold**.

Qianfan-OCR achieves the best scores on six out of eight benchmarks, with particular strength on chart and academic reasoning tasks: CharXiv_DQ (94.0), CharXiv_RQ (85.2), ChartQA (88.1), ChartQAPro (42.9), ChartBench (85.9), and OCRVQA (66.8). On general document understanding benchmarks, Qwen3-VL-4B outperforms Qianfan-OCR on DocVQA (94.9 vs. 92.8) and TextVQA (81.8 vs. 80.0). This gap is consistent with the trend observed in Section 6.2: as a model designed for specialized OCR tasks, Qianfan-OCR accepts modest performance differences on general understanding benchmarks relative to same-size general VLMs, while achieving substantially stronger results on domain-specific OCR benchmarks (OmniDocBench, OlmOCR) and chart-related tasks where structural visual reasoning is critical.

The most striking result is the complete failure of all two-stage systems on CharXiv (0.0 on both DQ and RQ), where chart structures, axis relationships, and data point positions – discarded during text extraction – are essential for answering. The degradation extends to other chart benchmarks (ChartQA: 7.0–56.8 vs. 88.1; ChartBench: 2.0–17.2 vs. 85.9) and even text-heavy tasks like DocVQA (55.9–67.1 vs. 92.8–94.9), confirming that spatial and layout context provides value beyond what plain text can capture.

6.4 Key Information Extraction Benchmarks

We evaluate KIE performance on five public benchmarks as described in Section 5. Since specialized OCR models lack native KIE capabilities, we compare Qianfan-OCR with commercial models (Gemini-3.1-Pro, Gemini-3-Pro, Seed-2.0) and open-source models (Qwen3-4B-VL, Qwen3-VL-235B-A22B).

Model	Overall (Mean)	OCRBench KIE	OCRBenchv2 KIE (en)	OCRBenchv2 KIE (zh)	CCOCR KIE	Nanonets KIE (F1)
Qianfan-OCR (Ours)	**87.9**	95.0	82.8	**82.3**	92.8	**86.5**
Qwen3-4B-VL	83.5	89.0	82.1	71.3	91.6	83.3
Qwen3-VL-235B-A22B	84.2	94.0	85.6	62.9	**95.1**	83.8
Gemini-3.1-Pro	79.2	**96.0**	**87.8**	63.4	72.5	76.1
Gemini-3-Pro	77.0	93.5	87.1	49.6	72.7	82.1
Seed-2.0	78.0	92.5	75.6	48.9	89.6	83.4

Table 7: Performance comparison on Key Information Extraction (KIE) benchmarks (normalized scores). Qianfan-OCR achieves the highest overall mean score of 87.9 across five public KIE benchmarks, outperforming both state-of-the-art commercial models and similarly-sized open-source models.

As shown in Table 7, Qianfan-OCR achieves the highest overall mean score of 87.9 across five public KIE benchmarks, outperforming the similarly-sized Qwen3-4B-VL (83.5) by 4.4 points, the much larger Qwen3-VL-235B-A22B (84.2) by 3.7 points, and commercial large models by approximately 9–11 points. Qianfan-OCR leads on OCRBenchv2 KIE Chinese (82.3) and Nanonets KIE F1 (86.5), demonstrating strong advantages on Chinese document extraction and real-world evaluation scenarios. Notably, even Qwen3-VL-235B-A22B with over 50× more activated parameters achieves a lower overall score, primarily due to weak Chinese KIE performance (62.9 on OCRBenchv2 KIE zh). The Gemini-3 series achieves strong results on English KIE tasks (Gemini-3.1-Pro: OCRBench KIE 96.0, OCRBenchv2 KIE en 87.8), but drops sharply on Chinese benchmarks (OCRBenchv2 KIE zh: 63.4 and 49.6), indicating limited multilingual generalization. Seed-2.0 shows relatively weaker performance on OCRBenchv2 KIE tasks (en 75.6, zh 48.9) compared to other models. These results validate the effectiveness of Qianfan-OCR's end-to-end architecture for spatial reasoning and field association tasks across both Chinese and English scenarios.

6.5 Inference Throughput

Beyond accuracy, inference throughput is a critical factor for production deployment. Since two-stage pipeline systems (e.g., PaddleOCR-VL) involve heterogeneous components beyond the language model (layout detection, element-wise recognition, post-processing), raw model throughput alone is not a meaningful comparison metric. We therefore adopt **pages per second (PPS)** – the number of complete document pages parsed per second – as a holistic throughput measure that captures end-to-end system efficiency. All benchmarks are conducted on the OmniDocBench v1.5 dataset using a single NVIDIA A100 GPU with vLLM 0.10.2, consistent with the inference framework version reported in the PaddleOCR-VL technical report.

Model	Batch Size	# Query	PPS
PaddleOCR-VL†	–	–	1.224
MinerU 2.5†	–	–	1.057
MonkeyOCR-pro-1.2B†	–	–	0.673
Dots OCR†	–	–	0.352
Qianfan-OCR (W16A16)	512	512	0.503
Qianfan-OCR (W8A8)	512	512	**1.024**

Table 8: Inference throughput comparison measured in pages per second (PPS) on OmniDocBench v1.5 with a single A100 GPU. †Results from PaddleOCR-VL technical report.

As shown in Table 8, despite having a 4B-parameter language model backbone – substantially larger than the detection and recognition modules in pipeline systems – Qianfan-OCR with W8A8 quantization achieves 1.024 PPS, comparable to PaddleOCR-VL (1.224 PPS) and exceeding MonkeyOCR-pro-1.2B (0.673 PPS) and Dots OCR (0.352 PPS).

This competitive throughput is attributable to two architectural advantages of the end-to-end approach:

- **GPU-centric computation:** Two-stage pipeline systems rely on CPU-based layout analysis (detection, NMS, rule-based assembly) as a prerequisite for GPU-based recognition. Under high concurrency, the CPU stage becomes a bottleneck that throttles GPU utilization – a problem that worsens with more powerful GPUs. Qianfan-OCR processes entire pages through the GPU with minimal CPU involvement, avoiding this bottleneck entirely.

- **Efficient batching:** End-to-end models accept whole-page images that can be resized to uniform dimensions, enabling large-batch GPU inference with well-aligned memory access patterns. In contrast, pipeline systems process variable numbers of cropped regions per page, leading to irregular batch sizes and fragmented GPU utilization.
- **Lower deployment complexity:** Pipeline systems such as PaddleOCR-VL require asynchronous orchestration of data loading, layout analysis, and LLM inference stages, demanding careful tuning of per-stage concurrency, queue depths, and resource allocation to achieve optimal throughput. Qianfan-OCR reduces this to a standard single-model serving problem (e.g., a single vLLM instance), significantly lowering deployment effort and performance tuning cost.

With AWQ quantization at W8A8 precision, Qianfan-OCR achieves a 2× speedup over the W16A16 baseline (1.024 vs. 0.503 PPS) with negligible accuracy degradation, making it a practical choice for high-throughput document processing pipelines.

7 Limitations and Future Work

Qianfan-OCR represents an early exploration of unified end-to-end document intelligence with several limitations that warrant further investigation.

Layout-as-Thought. The current Layout-as-Thought mechanism has only been validated on OmniDocBench v1.5 for document parsing, where it shows targeted benefits on structurally complex pages. Its effectiveness on other tasks – such as key information extraction, document QA, and chart understanding – remains unexplored. The current implementation generates layout bounding boxes, labels, and brief text descriptions in a relatively rigid format via supervised fine-tuning. Future work should integrate these layout elements more naturally into the reasoning process, allowing the model to flexibly invoke spatial reasoning when needed rather than producing a fixed-format layout dump. Reinforcement learning is a promising direction to achieve this: by optimizing layout generation based on downstream task rewards, the model can learn to produce task-adaptive layout reasoning that selectively emphasizes relevant structural information, ultimately strengthening reasoning capabilities across diverse document intelligence scenarios.

Performance Ceiling. As a pioneering attempt at end-to-end OCR, the ultimate performance ceiling of purely end-to-end architectures remains an open question – future work should systematically explore architectural innovations, training strategies, and data scaling laws to determine whether end-to-end models can fully match or surpass heavily optimized pipeline systems.

Deployment Efficiency. Although W8A8 quantization enables competitive throughput on GPU (Section 6.5), Qianfan-OCR's 4B parameter footprint limits deployment in resource-constrained environments such as edge devices and CPU-only servers – future work should explore knowledge distillation and pruning to develop compact variants (1B–2B parameters) suitable for broader deployment scenarios. Beyond these core challenges, Qianfan-OCR exhibits limitations in video OCR, 3D text on curved surfaces, and highly stylized artistic handwriting, presenting promising directions for extending the unified architecture.

8 Conclusion

We present Qianfan-OCR, a 4B-parameter end-to-end model that unifies text recognition, layout analysis, and semantic understanding within a single vision-language architecture. Our key contributions include: (1) achieving state-of-the-art results among end-to-end models on OmniDocBench v1.5 and OlmOCR Bench, demonstrating that end-to-end architectures can be competitive with pipeline systems on recognition accuracy; (2) introducing Layout-as-Thought, a mechanism that integrates layout reasoning as optional chain-of-thought, enabling the model to dynamically invoke structural analysis for complex documents; (3) providing empirical evidence that two-stage OCR+LLM pipelines degrade substantially on tasks requiring spatial and visual reasoning, with zero accuracy on chart interpretation benchmarks where layout information is essential.

These results suggest that for document intelligence tasks requiring joint visual and textual understanding, preserving visual context throughout the processing pipeline offers significant advantages over text-only intermediate representations.

The model is publicly accessible through Baidu AI Cloud Qianfan platform at https://github.com/baidubce/Qianfan-VL.

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A Application Showcases

This appendix presents visual examples showcasing Qianfan-OCR's capabilities across diverse OCR scenarios and real-world applications.

A.1 Document Parsing and Layout Analysis

C14 Sports BOSTON SUNDAY GLOBE SEPTEMBER 21, 2025

Basketball

No excuses for Cavaliers, Knicks in East race

GARY WASHBURN

The 2025-26 NBA season is about here with media days for some teams coming this week, while the Celtics will open up their much-anticipated camp without **Jayson Tatum** next week in Brighton.

Parity is the theme, especially in the Eastern Conference. The Knicks and Cavaliers are considered the favorites, and the remaining 13 teams are a crapshoot. Orlando and Detroit are on the rise. Boston and Indiana are slipping.

Miami's top scorer is going to miss some of the regular season. In Toronto, the hope is that **Brandon Ingram** is healthy and will join **Scottie Barnes** and **RJ Barrett** for a potential contender. Philadelphia is just trying to stay healthy. **Doc Rivers** may be coaching for his job in Milwaukee.

Atlanta will play with expectations after a busy offseason. Chicago just brought back **Josh Giddey** and is hoping for a playoff run.

Here is a preview of the East, with comments on each team, projected win totals and predicted order of finish:

Cavaliers (60 wins)

The Cavaliers led the East with 64 wins last season and were primed for a long playoff run until running into the Pacers, who knocked them flat and made them look like a disheveled team. Cleveland brought back the same crew, then added **Lonzo Ball** to relieve some of the pressure off **Darius Garland** at point guard. **Evan Mobley** and **Jarrett Allen** are back along with **Donovan Mitchell**. There are no excuses this year. They folded in that series, which reflected poorly on first-year coach **Kenny Atkinson**. **Max Strus** will be out for the first few months of the season, but this team still has enough to win a lot of games.

Knicks (55 wins)

The Knicks made significant offseason changes, which included firing coach **Tom Thibodeau** and replacing him with well-traveled **Mike Brown** to jump-start the offense. They also upgraded the bench by adding former Celtic **Guerschon Yabusele** and scoring guard **Jordan Clarkson**. Their success will come down to how well **Jalen Brunson**, **Karl-Anthony Towns**, and **Mikal Bridges** play in the clutch. Like the Cavaliers, the Knicks have no excuses. They are talented and deeper than before. The question is whether Brown's style will translate better than Thibodeau's. If it doesn't, the Knicks are back where they started.

Magic (49 wins)

May be the shakiest third seed in years because the Magic haven't proven they can consistently win games, score, and stay healthy. **Paolo Banchero** and **Franz Wagner** led the team and are both healthy after missing significant time last year with oblique injuries. Orlando added **Desmond Bane** from the Grizzlies for shooting. Defensively, the Magic could be the most physical and punishing team in the NBA; they don't need a prolific offense, but they do need to be better. Bane can't be the only premium shooter. Wagner has to be better from the 3-point line, as does **Anthony Black**, the new starting point guard. Too many questions in Orlando for it to compete with the Cavaliers and Knicks, but it will be better.

Pistons (49 wins)

The Pistons won't be sneaking up on anyone this season. How will they deal with the raised expectations? The good news is **Jaden Ivey** is back from his leg injury, but the Pistons also lost sharpshooter **Malik Beasley** because of his personal issues and replaced him with **Duncan Robinson**. And if **Isaiah Stewart** was healthy during Detroit's first-round series against the Knicks, they may have advanced. The rest of the core returns intact, including **Cade Cunningham**, **Jalen Duren**, and **Ausar Thompson**. A sneaky good move for Detroit is the addition of former Michigan standout **Caris LeVert**. The Pistons are talented, but are they prepared for the pressure of being a contender?

Hawks (47 wins)

Talk about no excuses. The Hawks retooled their roster this summer, adding former Celtic **Kristaps Porzingis** and ex-Timberwolves standout **Nickeil Alexander-Walker** to join **Trae Young** and a healthy **Jalen Johnson**. In play-in land for years, it's about time they take the next step. That depends on how well Young can mesh with Porzingis. Speaking of Porzingis, the good news for Atlanta is he made it through EuroBasket healthy and those energy issues he experienced in Boston are gone, but he'll definitely be load-managed because of his history. Atlanta is deep after a series of strong drafts, but can the Hawks deal with actual expectations?

FILE/MICHAEL CONROY/ASSOCIATED PRESS

Kenny Atkinson coached the Cavaliers to an Eastern Conference-best 64-18 mark in his first season in Cleveland.

76ers (47 wins)

One of the more mysterious teams in the NBA. **Joel Embiid** was supposed to come back this season healthy, but there are questions as to whether he'll be ready. **Paul George** has been more productive in the podcast game than on the floor. **Tyrese Maxey** was Philadelphia's best player last season and he's really the only sure thing, but it did nab the promising **VJ Edgecombe** and **Johni Broome** in the draft while **Jared McCain**, who was enjoying a sparkling rookie season before a knee injury, returns fully healthy. **Nick Nurse's** job may be on the line this season and the impatience is growing with Embiid.

Celtics (45 wins)

Don't sleep on them, even without Tatum. **Jaylen Brown**, **Derrick White**, **Payton Pritchard**, and **Anfernee Simons** are a good enough quartet to win a share of games, and this team will be scrappy and play harder than their opponents. The question is whether **Joe Mazzulla** can devise an offensive strategy that will accentuate the roster's strengths. Being so 3-point reliant won't work with the departures of Porzingis, **Jrue Holiday**, and **Al Horford**. Boston still has questions at center, where third-stringer **Neemias Queta** is likely going to be elevated to starter. If **Luka Garza**, **Josh Minott**, and **Hugo Gonzalez** can produce, the Celtics could be a sleeper.

Bucks (44 wins)

Any team with **Giannis Antetokounmpo** is going to compete, but the Bucks just don't have enough talent to be amongst the elite. The organization is trying to do anything to keep Antetokounmpo from requesting a trade, even if it means re-signing his brother after a year's absence. The addition of **Myles Turner** will help in the paint, but the rest of the roster is still the same one that was so inconsistent last season. Trading fan favorite **Khris Middleton** for **Kyle Kuzma** was an abject failure, but the hope is Kuzma — along with **Kevin Porter Jr.**, **Gary Trent Jr.**, and **Bobby Portis** — will help enough to turn the Bucks into a surprise contender.

Raptors (42 wins)

It's about time the Raptors made a breakthrough; they have enough talent to win their share of games. Relying on Ingram is a big risk, but when healthy, he's a smooth-scoring forward who can also rebound. Barrett seems to have found a home in his native Canada and is a solid second option behind Barnes, who has learned to shoot the three as well as in the paint. The Raptors could be a sleeper team with the right luck, and they definitely have enough young talent to make it difficult on better teams.

Heat (39 wins)

Tyler Herro is expected to miss at least the first month of the season following ankle surgery, which dampens the team's chances of getting off to a strong start. There is talent here with **Bam Adebayo**, **Kel'el Ware**, and **Andrew Wiggins**, and the club also brought back defensive ace **Davion Mitchell**. With no major free agent acquisitions, the Heat will have to rely on the savvy coaching of **Erik Spoelstra** and the defensive prowess of Ware, Adebayo, and Wiggins to win ugly. The addition of **Norman Powell** should boost the offense, but Miami is definitely in the same place as it was before, vying for a play-in spot.

Pacers (37 wins)

It's going to be a transition season without **Tyrese Haliburton** and Turner, but coach **Rick Carlisle** will figure out a way to keep the club competitive. Like Brown in Boston, this could be a showcase for **Pascal Siakam** to be the unquestioned No. 1 offensive option, but the Pacers will need more from **Aaron Nesmith**, **Andrew Nembhard**, and **Bennedict Mathurin**. And they are going to be prone to suffer from an NBA Finals hangover.

Bulls (34 wins)

As usual, the Bulls did little to improve their squad in free agency, with their biggest acquisition **Isaac Okoro** from Cleveland. Chicago will rely on youth with Giddey, athletic maven **Matas Buzelis**, and **Coby White** while preparing for next summer, when it can sign potentially two max players.

Hornets (28 wins)

Hard to tell if the Hornets took a step forward under first-year coach **Charles Lee**, because they were often injured and inconsistent. A team with **LaMelo Ball**, **Brandon Miller**, and **Miles Bridges** should be competitive on most nights, but the Hornets still don't know how to win, and Ball hasn't shown the ability to consistently lead a team. It will be another growth season in Charlotte, nothing more.

Wizards (24 wins)

The Wizards again field a crew of youngsters along with a couple of veterans — Middleton and **CJ McCollum** — playing out their contracts. The goal, like in Brooklyn, is to be a major free agent player in 2026. The good news is **Alex Sarr**, **Bub Carrington**, **Cam Whitmore**, and **Bilal Coulibaly** all look like long-term prospects. If the plan works out, the Wizards won't be down for much longer.

Nets (20 wins)

The Nets had five first-round picks in this year's draft and selected all five, making already a painfully young team even more youthful. (Brooklyn has eight players on its roster age 21 or younger.) This will be a showcase for **Michael Porter Jr.** as a No. 1 option. **Nic Claxton** is a defensive maven and **Cameron Thomas** will be playing for free agency after coming back on a qualifying offer. It's going to be another long season at Barclays.

FURTHERING THE LEAGUE

Silver expands Europe thoughts

There has been more discussion in recent months about the NBA's delving into Europe and potentially beginning a league overseas than local expansion to candidates Las Vegas and Seattle. What's more, while there was expectation that commissioner **Adam Silver** would announce expansion cities in July, he tabled the topic and focused more on Europe.

When asked again last week in New York, Silver reiterated that he's exploring local expansion and fought back against the idea that he favors more European exploration.

"I don't know why it seems that way to some people. It's not the case," he said. "I think there are independent people looking at potential domestic expansion and then others looking at opportunities in Europe, and I see them as completely independent opportunities. The fact is no new news to report on domestic expansion, but it's something we continue to look at. We did discuss it at the board meeting. I think that we've spent a fair amount of time on the economic models around expansion. We've gotten into more of, I'd say, a deeper dive than when I last addressed the media on it. Certainly now that we know what our media contracts are. That helps in doing the math, at least over the next decade."

While there was speculation that 2030 could be a target for 31st and 32nd teams, there is no timetable. And the selling of the Lakers ($10 billion) and Celtics ($6.1 billion) have perhaps complicated the price of a new club.

"Part of the difficulty in potentially assessing it is a sense of long-term value of the league, and a little bit, maybe it's a high-class problem, but as with some of the recent jumps in franchise valuations, that sort of creates some confusion in the marketplace about how you might even price an expansion franchise," Silver said. "So I'll only say it's something that we continue to actively look at."

Silver also stressed Europe's status as a "clean slate . . . in terms of what the opportunity might be." That despite there already being a professional league in Europe with historically strong and valuable teams. The NBA will have to negotiate with the EuroLeague, and possibly FIBA, in plans for sponsored clubs in cities such as Paris and Madrid.

"Discussions with the EuroLeague are ongoing, so yes, there is still room to further engage with them and find ways that we can integrate our operation in some way," Silver said. "I wouldn't say it's our preference to link with those well-known soccer clubs in Europe. I think that is one model, where particularly some of those incredible soccer brands in Europe have some of the biggest followings in the world behind their clubs, but also, as you well know, have basketball traditions as well. They may not be as well known for their basketball clubs, but it's not as if we're taking a soccer brand and saying, now go start a basketball club."

Layups

Former Celtics guard **Dennis Schröder** helped Germany to its first EuroBasket gold medal since 1993 and was named MVP of the tournament. Schröder just signed a three-year deal with the Kings, his 10th NBA team in a 13-year career. While Schröder has bounced around the NBA, he still remains a quality, sought-after point guard. And his helping Germany to the FIBA World Cup and the EuroBasket titles in two years has potentially catapulted him into Naismith Memorial Hall of Fame status when he retires. Schröder turned 32 this week and still has a handful of NBA and international years left. Remember, the Hall of Fame has international entries, meaning there are several inductees who were elected, such as former Celtic **Dino Radja**, based mainly on their international performance. Schröder has become a dominant player in FIBA and has joined **Franz** and **Mo Wagner** and ex-Celtic **Daniel Theis** to help resurrect German basketball after the retirement of **Dirk Nowitzki** . . . While teams are filling rosters with Exhibit 10 contracts and other training camp invites, the Celtics added **Hayden Gray** and **Ron Harper Jr.** on deals, and have one more slot remaining. With the club needing frontcourt help, an intriguing invite could be former first-round pick **Thomas Bryant**, who was last with the Pacers and played significant minutes in the NBA Finals. The Celtics are short in the middle with **Neemias Queta**, **Luka Garza**, and **Amari Williams** (two-way contract) as their bigs. There are also a handful of veteran free agents who have yet to sign with clubs — **Alec Burks**, **Markelle Fultz**, **Talen Horton-Tucker**, and **Garrison Mathews** . . . The Nets continue to make roster moves that may only have significance inside the front office. They acquired former Atlanta first-round pick **Kobe Bufkin**, a former Michigan standout who played in 27 games over two seasons, for cash considerations. To make room for Bufkin, the Nets waived former first-round pick **Keon Johnson**, who started 56 games for the rebuilding team last season, averaging 10.4 points. Johnson, a 6-foot-5-inch guard, is only 23 so there should be a market for his services. The Nets enter this season as one of the league's youngest teams, with no player older than 28-year-old **Haywood Highsmith**. Brooklyn is barely over the salary cap floor and $52 million under the first apron.

Gary Washburn is a Globe columnist. He can be reached at gary.washburn@globe.com. Follow him @GwashburnGlobe.

Figure 6: Newspaper document parsing with complex multi-column layout

Recognition Result: Qianfan-OCR accurately extracts the complete article with proper multi-column layout parsing:

Basketball

No excuses for Cavaliers, Knicks in East race

GARY WASHBURN

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Raptors (41 wins)

The Raptors surprised everyone by winning 43 games last season despite trading Pascal Siakam and OG Anunoby. They have a nice young core with Scottie Barnes, Immanuel Quickley, and RJ Barrett, but they need to make the next step toward contention. Gradey Dick showed promise as a rookie and should take a step forward this season.

Heat (41 wins)

The Heat lost Caleb Martin to free agency and Jimmy Butler is another year older, but they still have Bam Adebayo, Tyler Herro, and a solid supporting cast. Miami always finds a way to compete in the playoffs, but their regular season consistency remains questionable.

Pacers (40 wins)

The Pacers shocked the Cavaliers in the playoffs last season but lost key pieces like Bruce Brown and their bench depth took a hit. Tyrese Haliburton and Bennedict Mathurin are a dynamic duo, but questions remain about their defense and whether they can repeat last year's success.

Bulls (37 wins)

Chicago is stuck in mediocrity. DeMar DeRozan departed for Sacramento, leaving Zach LaVine and Nikola Vucevic as the primary options. The Bulls need to decide if they're rebuilding or competing, because this current roster isn't good enough for either path.

Hornets (29 wins)

LaMelo Ball is healthy, which is good news for Charlotte. They added forward Miles Bridges back after his suspension and drafted Brandon Miller. The Hornets have young talent but need to develop winning habits and stay healthy to take the next step.

Wizards (25 wins)

Washington is in full rebuild mode after trading Bradley Beal. Jordan Poole is the veteran leader now, with young players like Bilal Coulibaly and Deni Avdija trying to develop. The Wizards will struggle but are playing for the future, not this season.

Nets (22 wins)

Brooklyn cleaned house, trading Kevin Durant and Kyrie Irving for draft picks and young players. Mikal Bridges leads this young squad that includes Cameron Johnson and Nic Claxton. The Nets are tanking while maintaining respectability, hoping to land a franchise cornerstone in the draft.

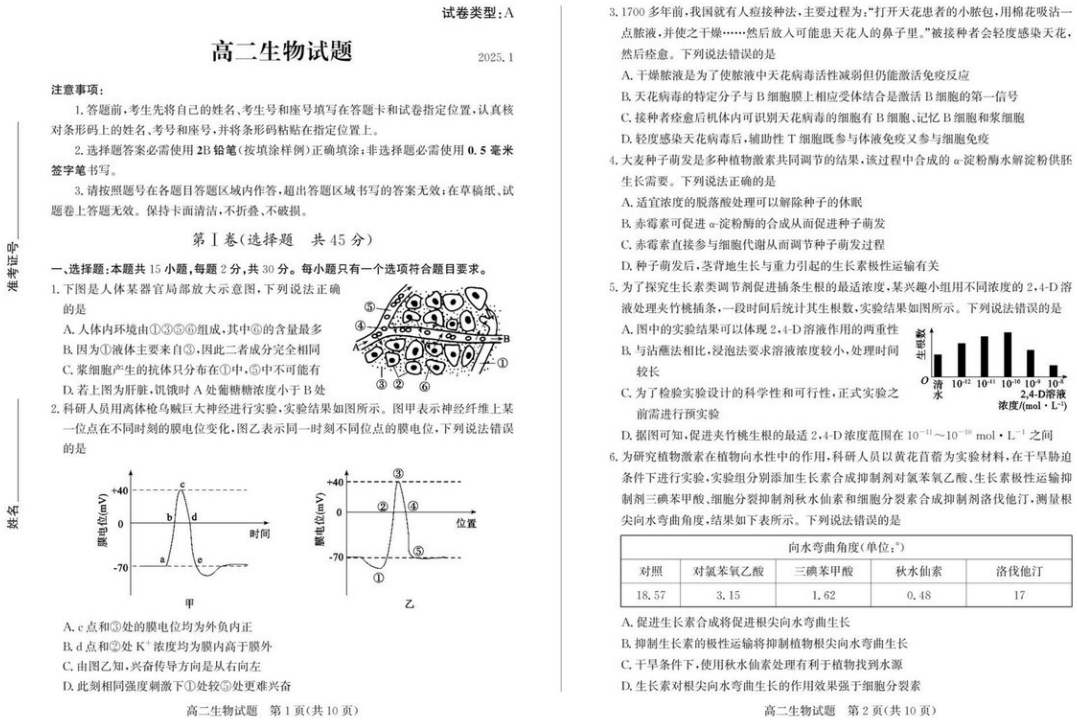
FURTHERING THE LEAGUE

Silver expands Europe thoughts

NBA commissioner Adam Silver confirmed the league is seriously exploring permanent expansion into Europe, with London and Paris as the primary candidates for potential franchises. Speaking at a press conference, Silver acknowledged the logistical challenges but emphasized the global growth strategy. "We've seen tremendous engagement in Europe, and our games there have been incredibly successful," Silver said. The league would need to address scheduling, travel concerns, and competitive balance before any final decision.

Layups

- The Lakers signed veteran guard Dennis Schroder to provide backcourt depth behind D'Angelo Russell.
- Former MVP Russell Westbrook announced his retirement after 16 seasons, finishing his career with the Clippers.
- The NBA approved new rules limiting transition take fouls, automatically awarding one free throw and possession to the offense.
- Several teams including the Suns and Mavericks face luxury tax penalties exceeding $50 million this season.
- International talent continues to surge, with over 120 players from outside the United States on opening night rosters.



试卷类型：A

高二生物试题

2025.1

注意事项：

1. 答题前，考生先将自己的姓名、考生号和座号填写在答题卡和试卷指定位置，认真核对条形码上的姓名、考号和座号，并将条形码粘贴在指定位置上。

2. 选择题答案必需使用 **2B 铅笔**（按填涂样例）正确填涂；非选择题必需使用 **0.5 毫米签字笔**书写。

3. 请按照题号在各题目答题区域内作答，超出答题区域书写的答案无效；在草稿纸、试题卷上答题无效。保持卡面清洁，不折叠、不破损。

准考证号

姓名

第Ⅰ卷（选择题　共 45 分）

一、选择题：本题共 15 小题，每题 2 分，共 30 分。每小题只有一个选项符合题目要求。

1. 下图是人体某器官局部放大示意图，下列说法正确的是

A. 人体内环境由①③⑤⑥组成，其中④的含量最多

B. 因为①液体主要来自③，因此二者成分完全相同

C. 浆细胞产生的抗体只分布在①中，⑤中不可能有

D. 若上图为肝脏，饥饿时 A 处葡萄糖浓度小于 B 处

2. 科研人员用离体枪乌贼巨大神经进行实验，实验结果如图所示。图甲表示神经纤维上某一位点在不同时刻的膜电位变化，图乙表示同一时刻不同位点的膜电位，下列说法错误的是

A. c 点和③处的膜电位均为外负内正

B. d 点和②处 K^+ 浓度均为膜内高于膜外

C. 由图乙知，兴奋传导方向是从右向左

D. 此刻相同强度刺激下①处较⑤处更难兴奋

3. 1700 多年前，我国就有人痘接种法，主要过程为："打开天花患者的小脓包，用棉花吸沾一点脓液，并使之干燥……然后放入可能患天花人的鼻子里。"被接种者会轻度感染天花，然后痊愈。下列说法错误的是

A. 干燥脓液是为了使脓液中天花病毒活性减弱但仍能激活免疫反应

B. 天花病毒的特定分子与 B 细胞膜上相应受体结合是激活 B 细胞的第一信号

C. 接种者痊愈后机体内可识别天花病毒的细胞有 B 细胞、记忆 B 细胞和浆细胞

D. 轻度感染天花病毒后，辅助性 T 细胞既参与体液免疫又参与细胞免疫

4. 大麦种子萌发是多种植物激素共同调节的结果，该过程中合成的 α-淀粉酶水解淀粉供胚生长需要。下列说法正确的是

A. 适宜浓度的脱落酸处理可以解除种子的休眠

B. 赤霉素可促进 α-淀粉酶的合成从而促进种子萌发

C. 赤霉素直接参与细胞代谢从而调节种子萌发过程

D. 种子萌发后，茎背地生长与重力引起的生长素极性运输有关

5. 为了探究生长素类调节剂促进插条生根的最适浓度，某兴趣小组用不同浓度的 2,4-D 溶液处理夹竹桃插条，一段时间后统计其生根数，实验结果如图所示。下列说法错误的是

A. 图中的实验结果可以体现 2,4-D 溶液作用的两重性

B. 与沾蘸法相比，浸泡法要求溶液浓度较小，处理时间较长

C. 为了检验实验设计的科学性和可行性，正式实验之前需进行预实验

D. 据图可知，促进夹竹桃生根的最适 2,4-D 浓度范围在 10^{-11}～10^{-10} mol·L^{-1} 之间

6. 为研究植物激素在植物向水性中的作用，科研人员以黄花苜蓿为实验材料，在干旱胁迫条件下进行实验，实验组分别添加生长素合成抑制剂对氯苯氧乙酸、生长素极性运输抑制剂三碘苯甲酸、细胞分裂抑制剂秋水仙素和细胞分裂素合成抑制剂洛伐他汀，测量根尖向水弯曲角度，结果如下表所示。下列说法错误的是

向水弯曲角度（单位：°）				
对照	对氯苯氧乙酸	三碘苯甲酸	秋水仙素	洛伐他汀
18.57	3.15	1.62	0.48	17

A. 促进生长素合成将促进根尖向水弯曲生长

B. 抑制生长素的极性运输将抑制植物根尖向水弯曲生长

C. 干旱条件下，使用秋水仙素处理有利于植物找到水源

D. 生长素对根尖向水弯曲生长的作用效果强于细胞分裂素

Figure 7: Exam paper OCR with mixed content (text, formulas, diagrams)

Recognition Result: Qianfan-OCR accurately recognizes the exam paper with mixed content including Chinese text, mathematical formulas, tables, and embedded figure annotations with bounding box coordinates:

准考证号姓名：____ 试卷类型：A

高二生物试题 2025.1

注意事项：

1. 答题前，考生先将自己的姓名、考生号和座号填写在答题卡和试卷指定位置，认真核对条形码上的姓名、考号和座号，并将条形码粘贴在指定位置上。
2. 选择题答案必需使用 2B 铅笔（按填涂样例）正确填涂；非选择题必需使用 0.5 毫米签字笔书写。
3. 请按照题号在各题目答题区域内作答，超出答题区域书写的答案无效；在草稿纸、试题卷上答题无效。保持卡面清洁，不折叠、不破损。

第 I 卷（选择题共 45 分）

一、选择题：本题共 15 小题，每题 2 分，共 30 分。每小题只有一个选项符合题目要求。

1. 下图是人体某器官局部放大示意图，下列说法正确的是
A. 人体内环境由 ①②③④ 组成，其中 ④ 的含量最多
B. 因为 ① 液体主要来自 ③，因此二者成分完全相同
C. 浆细胞产生的抗体只分布在 ① 中，⑤ 中不可能有
D. 若上图为肝脏，饥饿时 A 处葡糖糖浓度小于 B 处
![image](<box>[[<COORD_318>, <COORD_413>, <COORD_463>, <COORD_556>]]</box>)

2. 科研人员用离体枪乌贼巨大神经进行实验，实验结果如图所示。图甲表示神经纤维上某一位点在不同时刻的膜电位变化，图乙表示同一时刻不同位点的膜电位，下列说法错误的是
![image](<box>[[<COORD_102>, <COORD_655>, <COORD_260>, <COORD_823>]]</box>)甲
![image](<box>[[<COORD_285>, <COORD_653>, <COORD_438>, <COORD_823>]]</box>)乙
A. c 点和 ③ 处的膜电位均为外负内正
B. d 点和 ② 处 K^+ 浓度均为膜内高于膜外
C. 由图乙知，兴奋传导方向是从右向左
D. 此刻相同强度刺激下 ① 处较 ⑤ 处更难兴奋

3. 1700 多年前，我国就有人痘接种法，主要过程为："打开天花患者的小脓包，用棉花吸沾一点脓液，并使之干燥……然后放入可能患天花人的鼻子里。"被接种者会轻度感染天花，然后痊愈。下列说法错误的是
A. 干燥脓液是为了使脓液中天花病毒活性减弱但仍能激活免疫反应
B. 天花病毒的特定分子与 B 细胞膜上相应受体结合是激活 B 细胞的第一信号
C. 接种者痊愈后机体内可识别天花病毒的细胞有 B 细胞、记忆 B 细胞和浆细胞
D. 轻度感染天花病毒后，辅助性 T 细胞既参与体液免疫又参与细胞免疫

4. 大麦种子萌发是多种植物激素共同调节的结果，该过程中合成的 α -淀粉酶水解淀粉供胚生长需要。下列说法正确的是
A. 适宜浓度的脱落酸处理可以解除种子的休眠
B. 赤霉素可促进 α -淀粉酶的合成从而促进种子萌发
C. 赤霉素直接参与细胞代谢从而调节种子萌发过程
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C. 为了检验实验设计的科学性和可行性，正式实验之前需进行预实验
D. 据图可知，促进夹竹桃生根的最适 2,4-D 浓度范围在 $10^{-11} \sim 10^{-10}$ mol·L^{-1} 之间
![image](<box>[[<COORD_894>, <COORD_470>, <COORD_953>, <COORD_567>]]</box>)

6. 为研究植物激素在植物向水性中的作用，科研人员以黄花苜蓿为实验材料，在干旱胁迫条件下进行实验，实验组分别添加生长素合成抑制剂对氯苯氧乙酸、生长素极性运输抑制剂三碘苯甲酸、细胞分裂抑制剂秋水仙素和细胞分裂素合成抑制剂洛伐他汀，测量根尖向水弯曲角度，结果如下表所示。下列说法错误的是

向水弯曲角度（单位：°）	对照	氯苯氧乙酸	三碘苯甲酸	秋水仙素	洛伐他汀
	18.57	3.15	1.62	0.48	17

A. 促进生长素合成将促进根尖向水弯曲生长
B. 抑制生长素的极性运输将抑制植物根尖向水弯曲生长
C. 干旱条件下，使用秋水仙素处理有利于植物找到水源
D. 生长素对根尖向水弯曲生长的作用效果强于细胞分裂素

A.2 Table Recognition

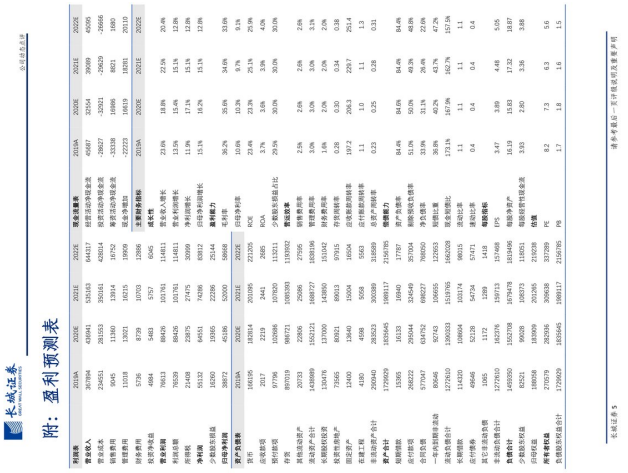


Figure 8: Rotated table recognition and structure parsing

Recognition Result: Qianfan-OCR recognizes the rotated financial tables with accurate structure preservation:

利润表				
	2019A	2020E	2021E	2022E
营业收入	367894	436941	535163	644317
营业成本	234551	281553	350161	428014
销售费用	9045	11360	13914	16752
管理费用	11018	13021	16215	19909
财务费用	5736	8739	10703	12886
投资净收益	4984	5483	5757	6045
营业利润	76613	88426	101761	114811
利润总额	76539	88426	101761	114811
所得税	21408	23875	27475	30999
净利润	55132	64551	74286	83812
少数股东损益	16260	19365	22286	25144
归母净利润	38872	45186	52000	58668

现金流量表				
经营活动净现金流	45687	32554	39089	45095
投资活动净现金流	-28627	-32921	-29629	-26666
筹资活动净现金流	-33338	16986	8821	1680
现金净增加	-22223	16619	18281	20110

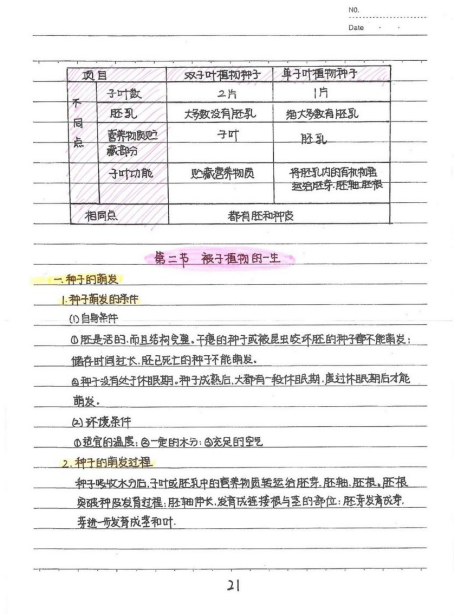
成长性/盈利能力				
营业收入增长	23.6%	18.8%	22.5%	20.4%
营业利润增长	13.5%	15.4%	15.1%	12.8%
归母净利润增长	15.1%	16.2%	15.1%	12.8%
毛利率	36.2%	35.6%	34.6%	33.6%
归母净利率	10.6%	10.3%	9.7%	9.1%
ROE	23.4%	23.3%	25.1%	25.9%
ROA	3.7%	3.6%	3.9%	4.0%

资产负债表				
	2019A	2020E	2021E	2022E
货币	166195	182814	201095	221205
应收款项	2017	2219	2441	2685
预付款项	97796	102686	107820	113211
存货	897019	986721	1085393	1193932
其他流动资产	20733	22806	25086	27595
流动资产合计	1438989	1552121	1688727	1838196
长期股权投资	130476	137000	143850	151042
投资性房地产	73565	80921	89013	97915
固定资产	12400	13640	15004	16504
在建工程	4180	4598	5058	5563
非流动资产合计	290940	283523	300389	318589
资产合计	1729929	1835645	1989117	2156785
短期借款	15365	16133	16940	17787
应付款项	268222	295044	324549	357004
合同负债	577047	634752	698227	768050
一年内到期非流动	80646	92743	106655	122653
流动负债合计	1272610	1390333	1519765	1662028
长期借款	114320	108604	103174	98015
应付债券	49646	52128	54734	57471
其它非流动负债	1065	1172	1289	1418
非流动负债合计	1272610	162376	159713	157468
负债合计	1459350	1552708	1679478	1819496
少数股东权益	82521	99028	108373	118051
归母权益	188058	183909	201265	219238
所有者权益	270579	282936	309638	337289
负债股东权益合计	1729929	1835645	1989117	2156785

主要财务指标				
少数股东损益占比	29.5%	30.0%	30.0%	30.0%
销售/管理/财务费用率	2.5/3.0/1.6	2.6/3.0/2.0	2.6/3.0/2.0	2.6/3.1/2.0
资产负债率	84.4%	84.6%	84.4%	84.4%
净负债率	33.9%	31.1%	26.4%	22.6%
流动/速动比率	1.1/0.4	1.1/0.4	1.1/0.4	1.1/0.4
EPS	3.47	3.89	4.48	5.05
PE / PB	8.2/1.7	7.3/1.8	6.3/1.6	5.6/1.5

Table 9: Financial forecast tables extracted from rotated document with accurate structure preservation

A.3 Handwriting Recognition



Recognition Result:

<table><tr><td colspan="2"> 项 目 </td><td> 双 子 叶 植 物 种 子 </td><td> 单 子 叶 植 物 种 子 </td></tr><tr><td rowspan="4"> 不 同 点 </td><td> 子 叶 数 </td><td>2 片 </td><td>1 片 </td></tr><tr><td> 胚 乳 </td><td> 大 多 数 没 有 胚 乳 </td><td> 绝 大 多 数 有 胚 乳 </td></tr><tr><td> 营 养 物 质 贮 藏 部 分 </td><td> 子 叶 </td><td> 胚 乳 </td></tr><tr><td> 子 叶 功 能 </td><td> 贮 藏 营 养 物 质 </td><td> 将 胚 乳 内 的 有 机 物 转 运 给 胚 芽、胚 轴、胚 根 </td></tr><tr><td colspan="2"> 相 同 点 </td><td colspan="2"> 都 有 胚 和 种 皮 </td></tr></table>

第二节被子植物的一生

一、种子的萌发

1. 种子萌发的条件

(1) 自身条件

① 胚是活的，而且结构完整。干瘪的种子或被昆虫咬坏胚的种子都不能萌发；储存时间过长，胚已死亡的种子不能萌发。

② 种子没有处于休眠期。种子成熟后，大都有一段休眠期，度过休眠期后才能萌发。

(2) 环境条件

① 适宜的温度；② 一定的水分；③ 充足的空气

2. 种子的萌发过程

种子吸收水分后，子叶或胚乳中的营养物质转运给胚芽、胚轴、胚根。胚根突破种皮发育过程：胚轴伸长，发育成连接根与茎的部位；胚芽发育成芽，芽进一步发育成茎和叶。

Figure 9: Handwritten text recognition

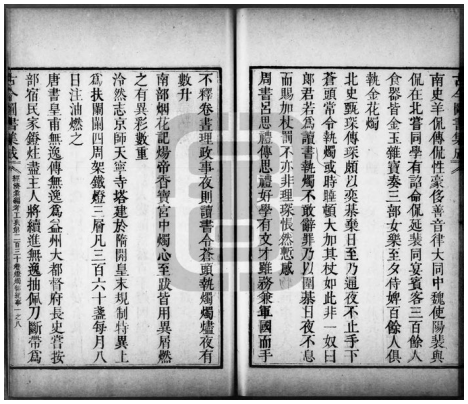


Figure 10: Vertical ancient Chinese text recognition

Recognition Result:

古今圖書集成

南史羊侃傳侃性豪侈善音律大同中魏使陽裴與侃在北嘗同學有F命侃延裴同宴賓客三百餘人食器皆金玉雜寶奏三部女樂至夕侍婢百余人俱執金花燭

北史甄琛傳琛頗以奕基F日至乃逼夜不止手下蒼頭常令執燭或時睡頓大加其杖如此非一奴曰郎君若F讀書執燭不敢辭罪乃以圍基日夜不息而賜加杖罰不亦非理琛F然F感

周書F思禮傳思禮好學有文才雖務兼軍國而手不釋卷晝理政事夜則讀書令蒼頭執燭爐夜有數升

南部烟花記F帝香寶F中燭心至跋皆用F屑燃之有F彩數重

冷然志京師天寧寺塔建於隋開皇末規制特F上F扶闌闌四周架鐵燈三層凡三百六十盞每月八日注油燃之

唐書皇甫無逸傳無逸F益州大都督府長史嘗按部宿民家F炷盡主人將續進無逸抽佩刀斷帶F

A.4 Key Information Extraction

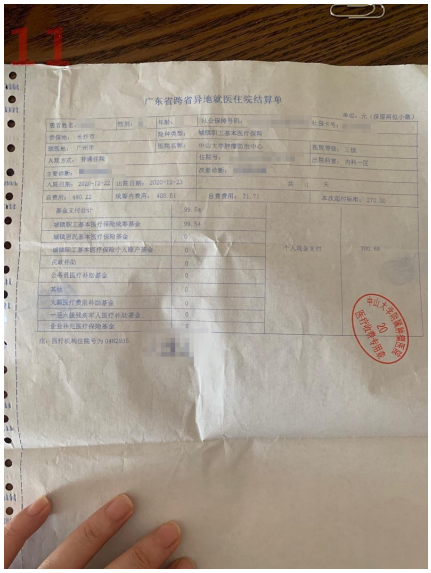


Figure 11: Medical paper key information extraction

User Prompt: 请从图片中提取以下字段信息：姓名、入院时间、出院时间、总费用、自费费用、医保支付。注意：字段名与图像中的原始文字含义一致，但不一定完全一样。仅提取清晰可见且可确定的文字内容。值完整提取及格式保持，使用标准 JSON 格式。

Extraction Result:

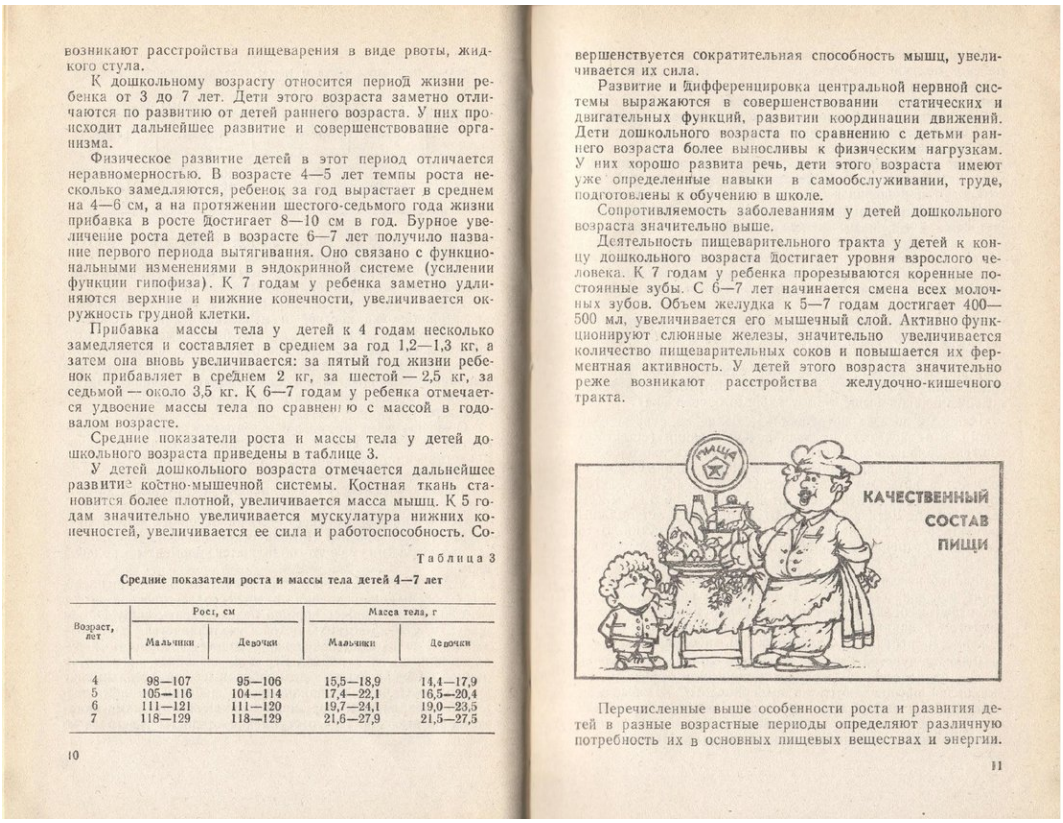
```
{"姓名": "", "入院时间": "2020-12-22",
 "出院时间": "2020-12-23", "总费用": "480.22",
 "自费费用": "71.71", "医保支付": "99.54"}
```

User Prompt: 请以 json 格式输出图中的 key-value
Extraction Result:


```
{"白细胞计数": {"值":"7.70","参考范围":"3.5-9.5","单位":"*10~9/L"},
 "红细胞计数": {"值":"4.51","参考范围":"3.8-5.1","单位":"*10~12/L"},
 "血红蛋白": {"值":"133","参考范围":"115-150","单位":"g/L"},
 "红细胞压积": {"值":"39.9","参考范围":"35-45","单位":"%"},
 "平均红细胞体积": {"值":"88.5","参考范围":"82-100","单位":"fL"},
 "平均血红蛋白量": {"值":"29.5","参考范围":"27-34","单位":"pg"},
 "平均血红蛋白浓度": {"值":"333","参考范围":"316-354","单位":"g/L"},
 "红细胞体积分布宽度-SD":{"值":"38.4","参考范围":"37-51","单位":"fL"},
 "红细胞体积分布宽度-CV":{"值":"11.9","参考范围":"0-14.9","单位":"%"},
 "血小板计数": {"值":"260","参考范围":"125-350","单位":"*10~9/L"}}
```


Figure 12: Mobile phone specification extraction

A.5 Multilingual OCR



возникают расстройства пищеварения в виде рвоты, жидкого стула.

К дошкольному возрасту относится период жизни ребенка от 3 до 7 лет. Дети этого возраста заметно отличаются по развитию от детей раннего возраста. У них происходит дальнейшее развитие и совершенствование организма.

Физическое развитие детей в этот период отличается неравномерностью. В возрасте 4—5 лет темпы роста несколько замедляются, ребенок за год вырастает в среднем на 4—6 см, а на протяжении шестого-седьмого года жизни прибавка в росте достигает 8—10 см в год. Бурное увеличение роста детей в возрасте 6—7 лет получило название первого периода вытягивания. Оно связано с функциональными изменениями в эндокринной системе (усилении функции гипофиза). К 7 годам у ребенка заметно удлиняются верхние и нижние конечности, увеличивается окружность грудной клетки.

Прибавка массы тела у детей к 4 годам несколько замедляется и составляет в среднем за год 1,2—1,3 кг, а затем она вновь увеличивается: за пятый год жизни ребенок прибавляет в среднем 2 кг, за шестой — 2,5 кг, за седьмой — около 3,5 кг. К 6—7 годам у ребенка отмечается удвоение массы тела по сравнению с массой в годовалом возрасте.

Средние показатели роста и массы тела у детей дошкольного возраста приведены в таблице 3.

У детей дошкольного возраста отмечается дальнейшее развитие костно-мышечной системы. Костная ткань становится более плотной, увеличивается масса мышц. К 5 годам значительно увеличивается мускулатура нижних конечностей, увеличивается ее сила и работоспособность. Совершенствуется сократительная способность мышц, увеличивается их сила.

Таблица 3

Средние показатели роста и массы тела детей 4—7 лет

Возраст, лет	Рост, см		Масса тела, г	
	Мальчики	Девочки	Мальчики	Девочки
4	98—107	95—106	15,5—18,9	14,4—17,9
5	105—116	104—114	17,4—22,1	16,5—20,4
6	111—121	111—120	19,7—24,1	19,0—23,5
7	118—129	118—129	21,6—27,9	21,5—27,5

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Развитие и дифференцировка центральной нервной системы выражаются в совершенствовании статических и двигательных функций, развитии координации движений. Дети дошкольного возраста по сравнению с детьми раннего возраста более выносливы к физическим нагрузкам. У них хорошо развита речь, дети этого возраста имеют уже определенные навыки в самообслуживании, труде, подготовлены к обучению в школе.

Сопротивляемость заболеваниям у детей дошкольного возраста значительно выше.

Деятельность пищеварительного тракта у детей к концу дошкольного возраста достигает уровня взрослого человека. К 7 годам у ребенка прорезываются коренные постоянные зубы. С 6—7 лет начинается смена всех молочных зубов. Объем желудка к 5—7 годам достигает 400—500 мл, увеличивается его мышечный слой. Активно функционируют слюнные железы, значительно увеличивается количество пищеварительных соков и повышается их ферментная активность. У детей этого возраста значительно реже возникают расстройства желудочно-кишечного тракта.

Перечисленные выше особенности роста и развития детей в разные возрастные периоды определяют различную потребность их в основных пищевых веществах и энергии.

11

Figure 13: Multilingual book with figures and mixed scripts

Recognition Result: Qianfan-OCR accurately extracts multilingual text (Russian), tables, and figure annotations from educational materials:

возникают расстройства пищеварения в виде рвоты, жидкого стула.К дошкольному возрасту относится период жизни ребенка от 3 до 7 лет. Дети этого возраста заметно отличаются по развитию от детей раннего возраста. У них происходит дальнейшее развитие и совершенствование организма.

Физическое развитие детей в этот период отличается неравномерностью. В возрасте 4—5 лет темпы роста несколько замедляются, ребенок за год вырастает в среднем на 4—6 см, а на протяжении шестого-седьмого года жизни прибавка в росте достигает 8—10 см в год. Бурное увеличение роста детей в возрасте 6—7 лет получило название первого периода вытягивания. Оно связано с функциональными изменениями в эндокринной системе (усиление функции гипофиза). К 7 годам у ребенка заметно удлиняются верхние и нижние конечности, увеличивается окружность грудной клетки.

Прибавка массы тела у детей к 4 годам несколько замедляется и составляет в среднем за год 1,2—1,3 кг, а затем она вновь увеличивается: за пятый год жизни ребенок прибавляет в среднем 2 кг, за шестой —2,5 кг, за седьмой —около 3,5 кг. К 6—7 годам у ребенка отмечается удвоение массы тела по сравнению с массой в годовалом возрасте.

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Таблица 3
Средние показатели роста и массы тела детей 4—7 лет

<table><tr><td rowspan="2">Возраст, лет</td><td colspan="2">Рост, см</td><td colspan="2">Масса тела, г</td></tr><tr><td>Мальчики</td><td>Девочки</td><td>Мальчики</td><td>Девочки</td></tr><tr><td>4</td><td>98—107</td><td>95—106</td><td>15,5—18,9</td><td>14,4—17,9</td></tr><tr><td>5</td><td>105—116</td><td>104—114</td><td>17,4—22,1</td><td>16,5—20,4</td></tr><tr><td>6</td><td>111—121</td><td>111—120</td><td>19,7—24,1</td><td>19,0—23,5</td></tr><tr><td>7</td><td>118—129</td><td>118—129</td><td>21,6—27,9</td><td>21,5—27,5</td></tr></table>

вершенствуется сократительная способность мышц, увеличивается их сила.Развитие и дифференцировка центральной нервной системы выражаются в совершенствовании статических и двигательных функций, развитии координации движений. Дети дошкольного возраста по сравнению с детьми раннего возраста более выносливы к физическим нагрузкам. У них хорошо развита речь, дети этого возраста имеют уже определенные навыки в самообслуживании, труде, подготовлены к обучению в школе.

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![image](<box>[[<COORD_496>, <COORD_520>, <COORD_868>, <COORD_827>]]</box>)

Перечисленные выше особенности роста и развития детей в разные возрастные периоды определяют различную потребность их в основных пищевых веществах и энергии.

Recognition Result:

```
Brandenburger Tor
Denkmal für die ermordeten Juden Europas
700m   350m
Denkmal 17. Juni 1953
Preußisches Herrenhaus/ Bundesrat
300m   650m   250m
Potsdamer Platz
Leipz
13   12 - 20
by Wally-Jane Vignyngde
```

Figure 14: Multilingual street view OCR

Figure 15: Multilingual menu recognition

Recognition Result:

```
Menú
ENTRANTES
Gazpacho 3,00€ / Salmorejo 3,00€
Sopa de mariscos 5,9€
Ensalada mixta 5,9€ / César 5,9€
TAPAS
Jamón ibérico 12,00€
Queso manchego 4,90€
Calamares 6,00€ / Croquetas 2,70€
Pulpo a la gallega 7,50€
Huevos rotos 4,00€
Revuelto de espárragos 5,20€
Rabo de cerdo 3,90€
Patatas bravas 3,50€
Tortilla española 3,50€
Chipirones 6,00€
BEBIDAS
Agua 2,50€ / Refrescos 3,50€
Vino blanco Albariño 10,20€
Vino rosado R. del Duero 11,00€
Vino tinto Rioja 14,00€
ESPECIALIDADES
Paella mixta/mariscos 9,00€
Cocido madrileño 9,00€
Fabada asturiana 9,00€
CARNES
Pechugas de pollo 6,90€
Chuletas de cerdo 7,50€
Solomillo ibérico 10,90€
Entrecot de ternera 13,90€
Pluma ibérica 13,90€
PESCADOS
Bacalao rebozado 18,00€
Merluza en salsa 8,90€
Sepia a la plancha 10,90€
Lubina al horno 19,00€
Dorada a la sal 19,00€
POSTRES
Tarta de limón 2,90€
Bizcocho de chocolate 3,50€
Crema catalana 3,50€
Fruta 2,50€
Miguelitos de la Roda 3,50€
```

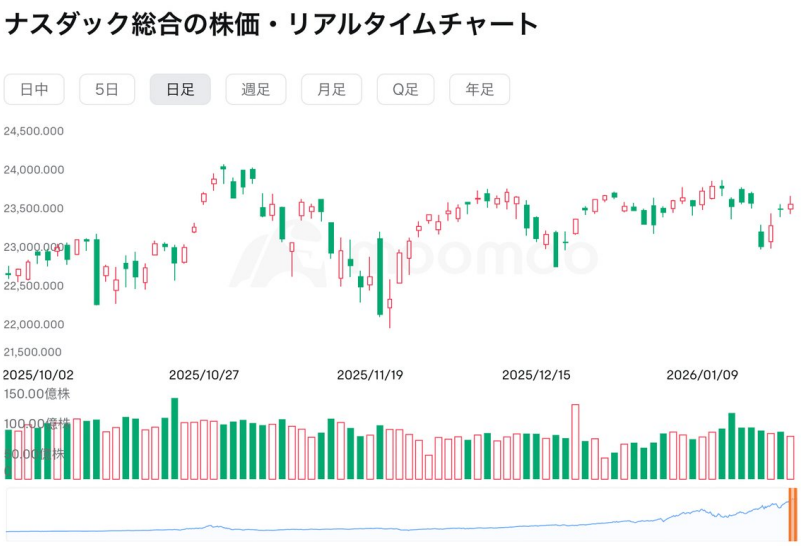

Figure 16: Multilingual chart understanding and analysis

User Prompt: 解读一下这张图片

Model Response:

这张图片是一个股票图表，显示了纳斯达克综合指数的股价走势。图表使用了蜡烛图（K 线图）的形式，绿色和红色的蜡烛分别代表股价的上涨和下跌。图表的顶部有日中、5 日、日足、Ⓔ足、月足、Q 足、年足等时间周期的选项，当前选择的是日足（日线图）。图表的左侧是价格轴，从 21,500.000 到 24,500.000，底部是日期轴，从 2025 年 10 月 2 日到 2026 年 1 月 9 日。图表下方的柱状图表示成交量，绿色和红色分别对应股价的上涨和下跌。整体来看，这段时间内纳斯达克综合指数的股价波动较大，但总体趋势是先上涨后下跌，最近又有所回升。

A.6 Formula Recognition

Musterlösung zu Blatt 1

Analysis I für Lehramt Gymnasium, Wintersemester 2007/08

1 a) Negation: Es gibt eine Ziehung der Lottozahlen, bei der niemand eine Million Euro gewinnt.
Die Aussage ist wohl falsch.

b) Negation: $\exists x \in \mathbb{R}\, \forall y \in \mathbb{R}\colon\ x \neq y^2$.
Die Aussage ist falsch, denn sie gilt nicht für negative $x \in \mathbb{R}$.

c) Negation: $\exists \varepsilon > 0\, \forall \delta > 0\, \exists x \in \mathbb{R}\colon\ |x| < \delta \ \wedge\ x^2 \geq \varepsilon$.
Die Aussage ist wahr, denn zu gegebenem $\varepsilon > 0$ läßt sich $\delta := \sqrt{\varepsilon} > 0$ wählen (Bedeutung der Aussage: die Funktion $x \mapsto x^2$ ist im Nullpunkt stetig).

2 a)
$$\begin{aligned} x \in (A \cap B) \cup C \ &\Leftrightarrow\ x \in A \cap B \ \vee\ x \in C \\ &\Leftrightarrow\ (x \in A \ \wedge\ x \in B) \ \vee\ x \in C \\ &\Leftrightarrow\ (x \in A \ \vee\ x \in C) \ \wedge\ (x \in B \ \vee\ x \in C) \\ &\Leftrightarrow\ (x \in A \cup C) \ \wedge\ (x \in B \cup C) \\ &\Leftrightarrow\ x \in (A \cup C) \cap (B \cup C) \end{aligned}$$

b)
$$\begin{aligned} x \in C \backslash (A \cap B) \ &\Leftrightarrow\ x \in C \ \wedge\ x \notin A \cap B \\ &\Leftrightarrow\ x \in C \ \wedge\ (x \notin A \ \vee\ x \notin B) \\ &\Leftrightarrow\ (x \in C \ \wedge\ x \notin A) \ \vee\ (x \in C \ \wedge\ x \notin B) \\ &\Leftrightarrow\ (x \in C \backslash A) \ \vee\ (x \in C \backslash B) \\ &\Leftrightarrow\ x \in (C \backslash A) \cup (C \backslash B) \end{aligned}$$

3 a) Die Aussage ist wahr, denn sind m und n gerade, so existieren $r, s \in \mathbb{N}$ mit $m = 2r$ und $n = 2s$. Dann ist $m \cdot n = 4rs = 2 \cdot 2rs$ mit $2rs \in \mathbb{N}$ und somit gerade.

b) Die Aussage ist falsch, denn sie gilt zum Beispiel nicht für $m = 2$ und $n = 1$.

c) Die Aussage ist wahr, denn:

Annahme: $\forall k \in \mathbb{N}\colon\ n \neq 2k$

Dann ist $n = 2k - 1$ für ein $k \in \mathbb{N}$ und somit
$$n^3 \ =\ (2k-1)^3 \ =\ 8k^3 - 12k^2 + 6k - 1 \ =\ 2(4k^3 - 6k^2 + 3k) - 1$$
mit $4k^3 - 6k^2 + 3k \in \mathbb{N}$, d.h. n^3 ist ungerade. **Widerspruch!**

Also ist $n = 2k$ für ein $k \in \mathbb{N}$ und somit gerade.

4 a) **1. Fall:** $x \geq 0$:
$$x + 1 \ \leq\ 2x \ \leq\ x + 2 \quad \Leftrightarrow \quad 1 \ \leq\ x \ \leq\ 2$$
2. Fall: $x < 0$:
$$x + 1 \ \leq\ -2x \ \leq\ x + 2 \quad \Leftrightarrow \quad 1 \ \leq\ -3x \ \leq\ 2 \quad \Leftrightarrow \quad -\frac{1}{3} \ \geq\ x \ \geq\ -\frac{2}{3}$$
Die Lösungsmenge ist also $[-\frac{2}{3}, -\frac{1}{3}] \cup [1, 2]$.

1

Figure 17: Mathematical formula recognition in multilingual context

Recognition Result: Qianfan-OCR accurately extracts German mathematical text with complex formulas and logical notation:

Musterlösung zu Blatt 1

Analysis I für Lehramt Gymnasium, Wintersemester 2007/08

1 a) Negation: Es gibt eine Ziehung der Lottozahlen, bei der niemand eine Million Euro gewinnt. Die Aussage ist wohl falsch.

b) Negation: $\exists x \in \mathbb{R} \forall y \in \mathbb{R} : x \neq y^2$. Die Aussage ist falsch, denn sie gilt nicht für negative $x \in \mathbb{R}$.

c) Negation: $\exists \varepsilon > 0 \forall \delta > 0 \exists x \in \mathbb{R} : |x| < \delta \wedge x^2 \geq \varepsilon$. Die Aussage ist wahr, denn zu gegebenem $\varepsilon > 0$ läßt sich $\delta := \sqrt{\varepsilon} > 0$ wählen (Bedeutung der Aussage: die Funktion $x \mapsto x^2$ ist im Nullpunkt stetig).

2 a)
$$\begin{aligned} x \in (A \cap B) \cup C \Leftrightarrow\ & x \in A \cap B \vee x \in C \\ \Leftrightarrow\ & (x \in A \wedge x \in B) \vee x \in C \\ \Leftrightarrow\ & (x \in A \vee x \in C) \wedge (x \in B \vee x \in C) \\ \Leftrightarrow\ & (x \in A \cup C) \wedge (x \in B \cup C) \\ \Leftrightarrow\ & x \in (A \cup C) \cap (B \cup C) \end{aligned}$$

b)
$$\begin{aligned} x \in C \backslash (A \cap B) \Leftrightarrow\ & x \in C \wedge x \notin A \cap B \\ \Leftrightarrow\ & x \in C \wedge (x \notin A \vee x \notin B) \\ \Leftrightarrow\ & (x \in C \wedge x \notin A) \vee (x \in C \wedge x \notin B) \\ \Leftrightarrow\ & (x \in C \backslash A) \vee (x \in C \backslash B) \\ \Leftrightarrow\ & x \in (C \backslash A) \cup (C \backslash B) \end{aligned}$$

3 a) Die Aussage ist wahr, denn sind m und n gerade, so existieren $r, s \in \mathbb{N}$ mit m = 2r und n = 2s. Dann ist $m \cdot n = 4rs = 2 \cdot 2rs$ mit $2rs \in \mathbb{N}$ und somit gerade.

b) Die Aussage ist falsch, denn sie gilt zum Beispiel nicht für m = 2 und n = 1.

c) Die Aussage ist wahr, denn:

Annahme: $\forall k \in \mathbb{N} : n \neq 2k$

Dann ist n = 2k - 1 für ein $k \in \mathbb{N}$ und somit

$$n^3 = (2k-1)^3 = 8k^3 - 12k^2 + 6k - 1 = 2(4k^3 - 6k^2 + 3k) - 1$$

mit $4k^3 - 6k^2 + 3k \in \mathbb{N}$, d.h. n^3 ist ungerade. Widerspruch!

Also ist n = 2k für ein $k \in \mathbb{N}$ und somit gerade.

4 a) 1. Fall: $x \geq 0$:

$$x + 1 \leq 2x \leq x + 2 \Leftrightarrow 1 \leq x \leq 2$$

2. Fall: x < 0:

$$x + 1 \leq -2x \leq x + 2 \Leftrightarrow 1 \leq -3x \leq 2 \Leftrightarrow -\frac{1}{3} \geq x \geq -\frac{2}{3}$$

Die Lösungsmenge ist also $\left[-\frac{2}{3}, -\frac{1}{3}\right] \cup [1, 2]$.